

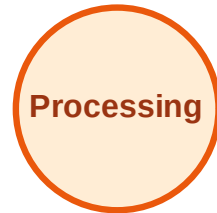
BFS Traversal

Step 1: Start at A

Legend



Complete



Processing



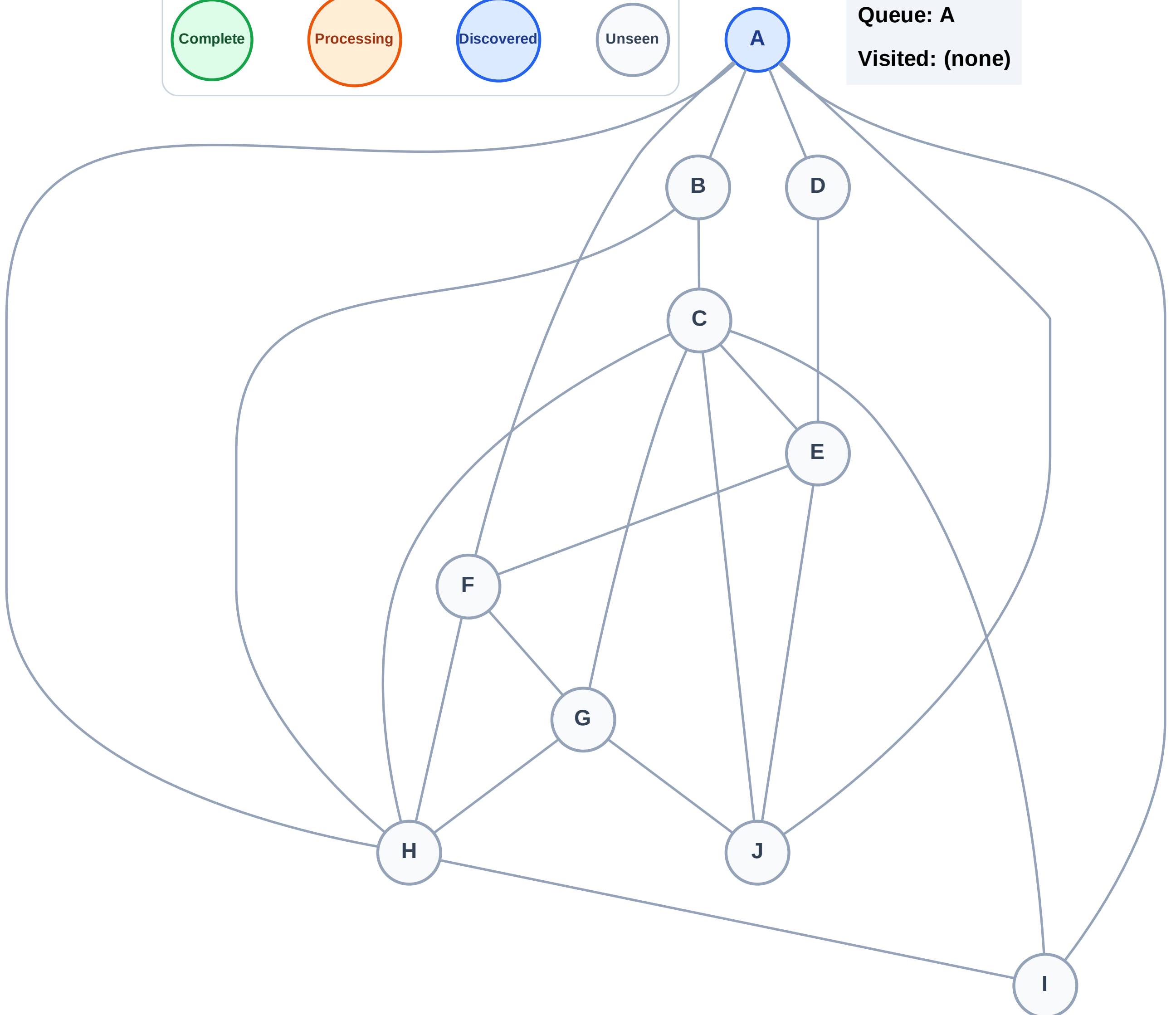
Discovered



Unseen

Queue: A

Visited: (none)



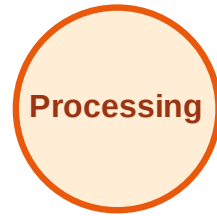
BFS Traversal

Step 2: Process node A

Legend



Complete



Processing



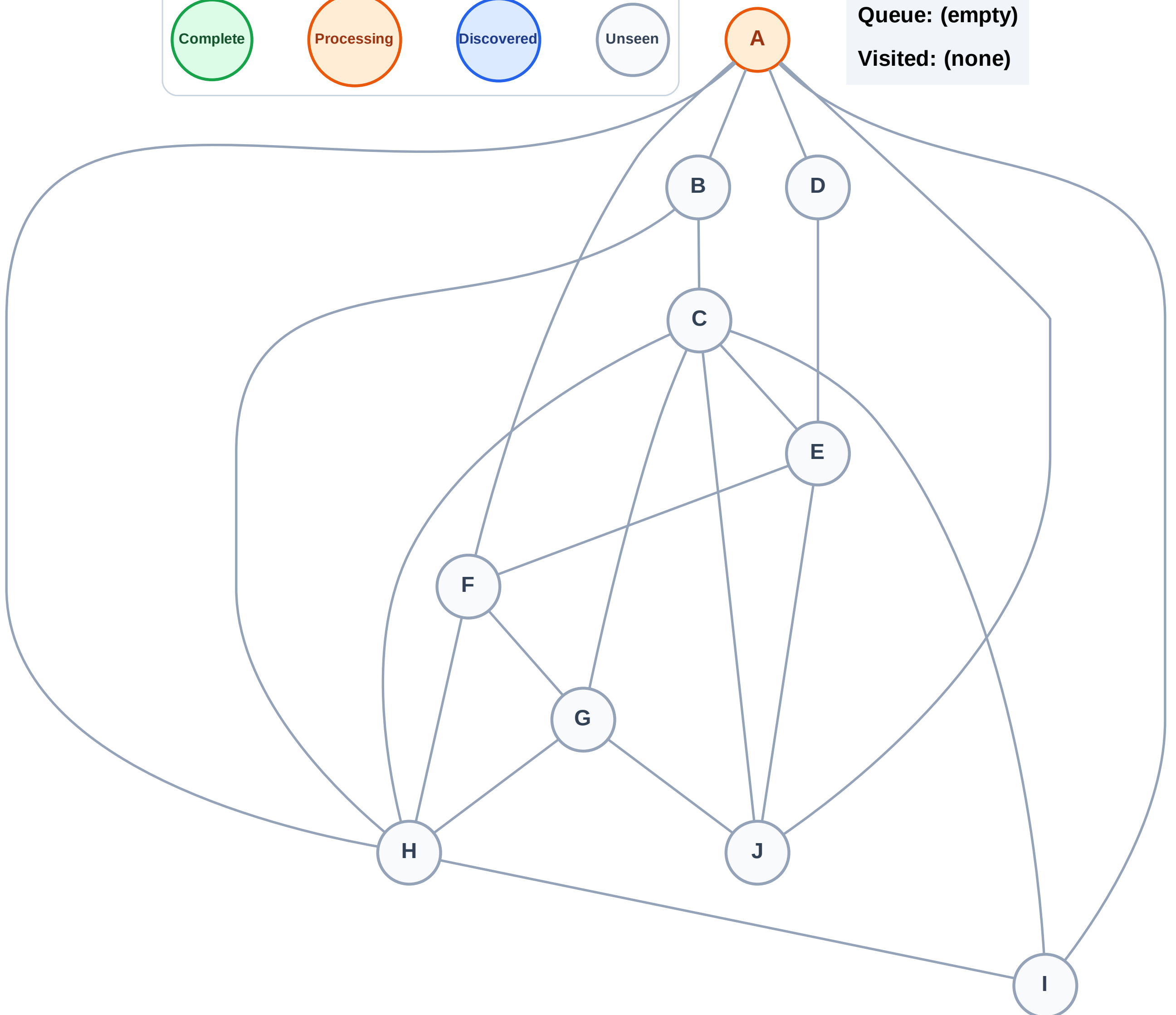
Discovered



Unseen

Queue: (empty)

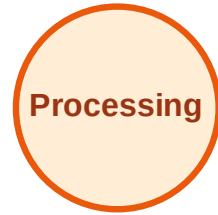
Visited: (none)



BFS Traversal

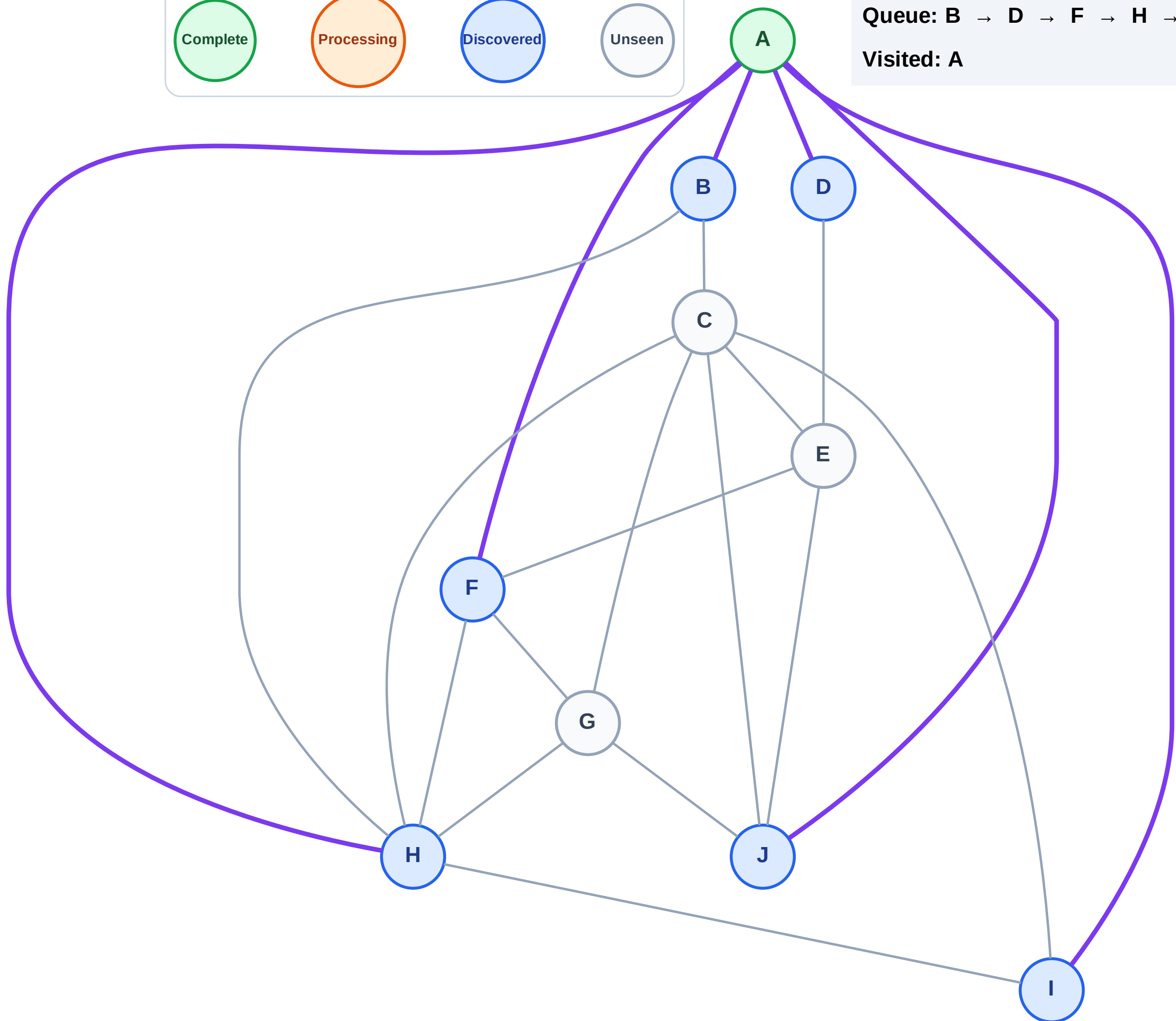
Step 3: Discover B, D, F, H, I, J from A

Legend



Queue: B → D → F → H → I → J

Visited: A



BFS Traversal

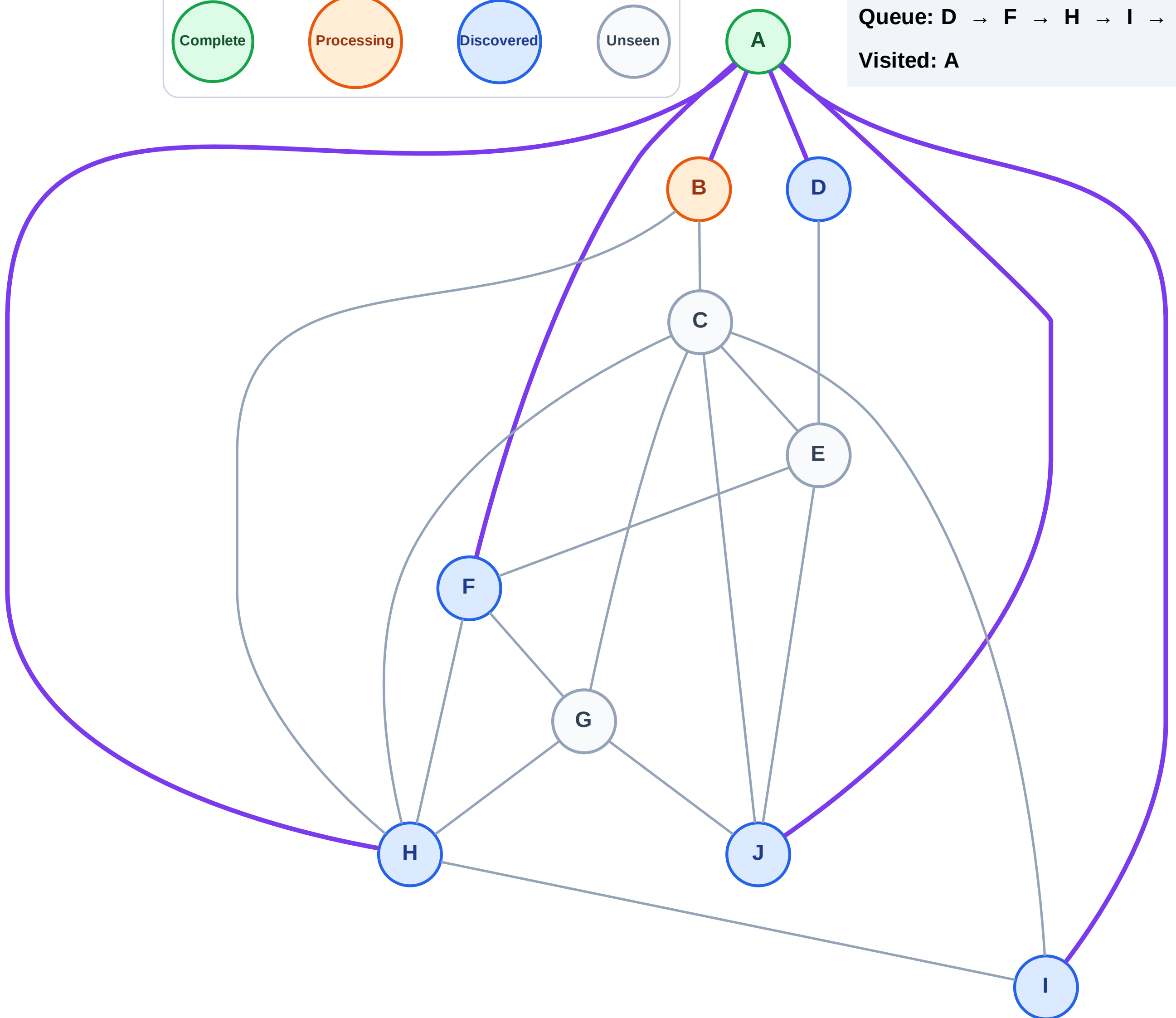
Step 4: Process node B

Legend



Queue: D → F → H → I → J

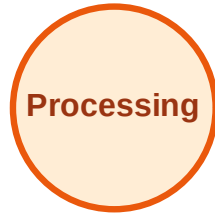
Visited: A



BFS Traversal

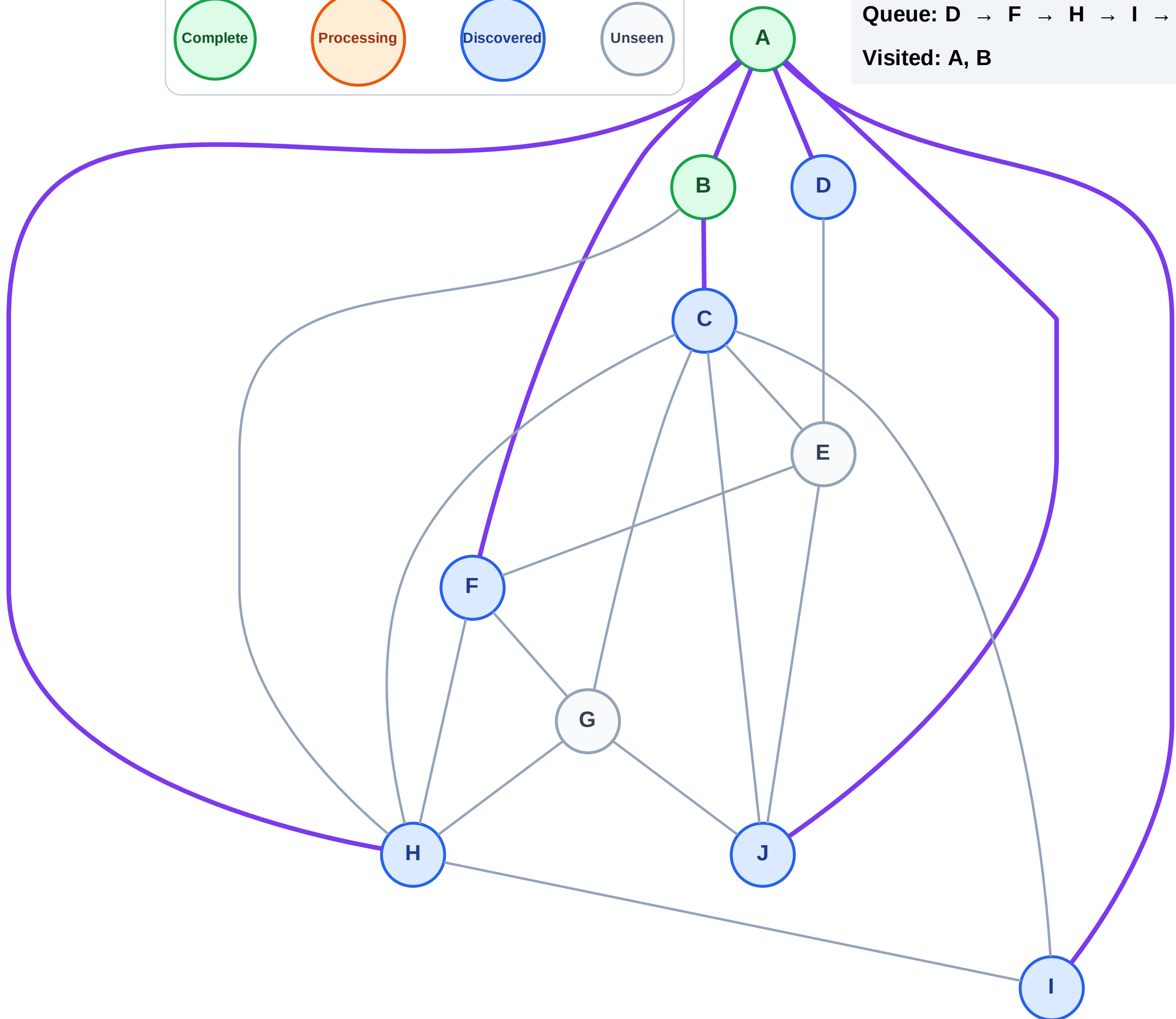
Step 5: Discover C from B

Legend



Queue: D → F → H → I → J → C

Visited: A, B



BFS Traversal

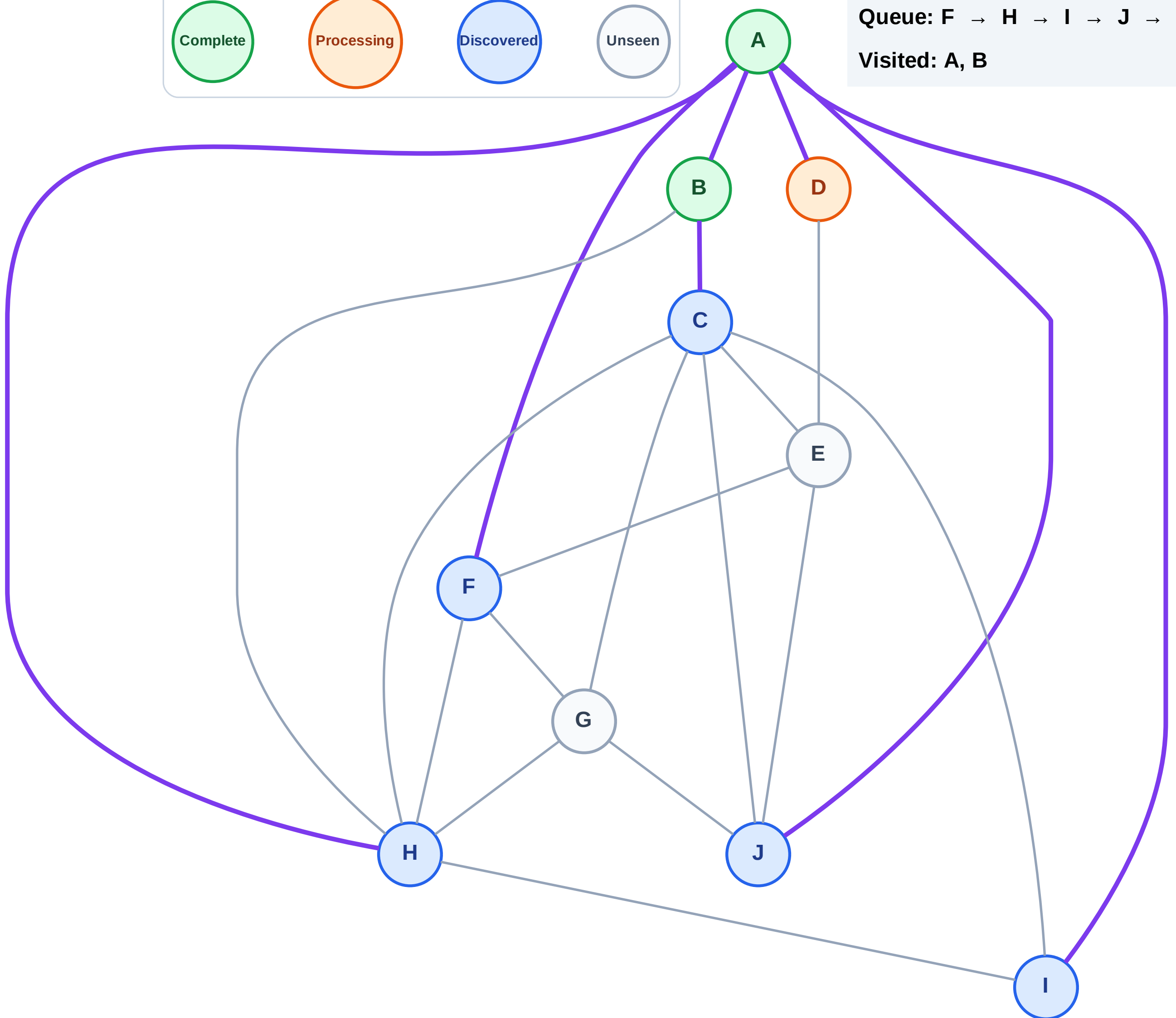
Step 6: Process node D

Legend



Queue: F → H → I → J → C

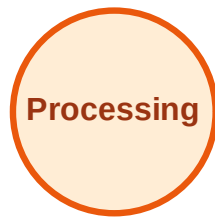
Visited: A, B



BFS Traversal

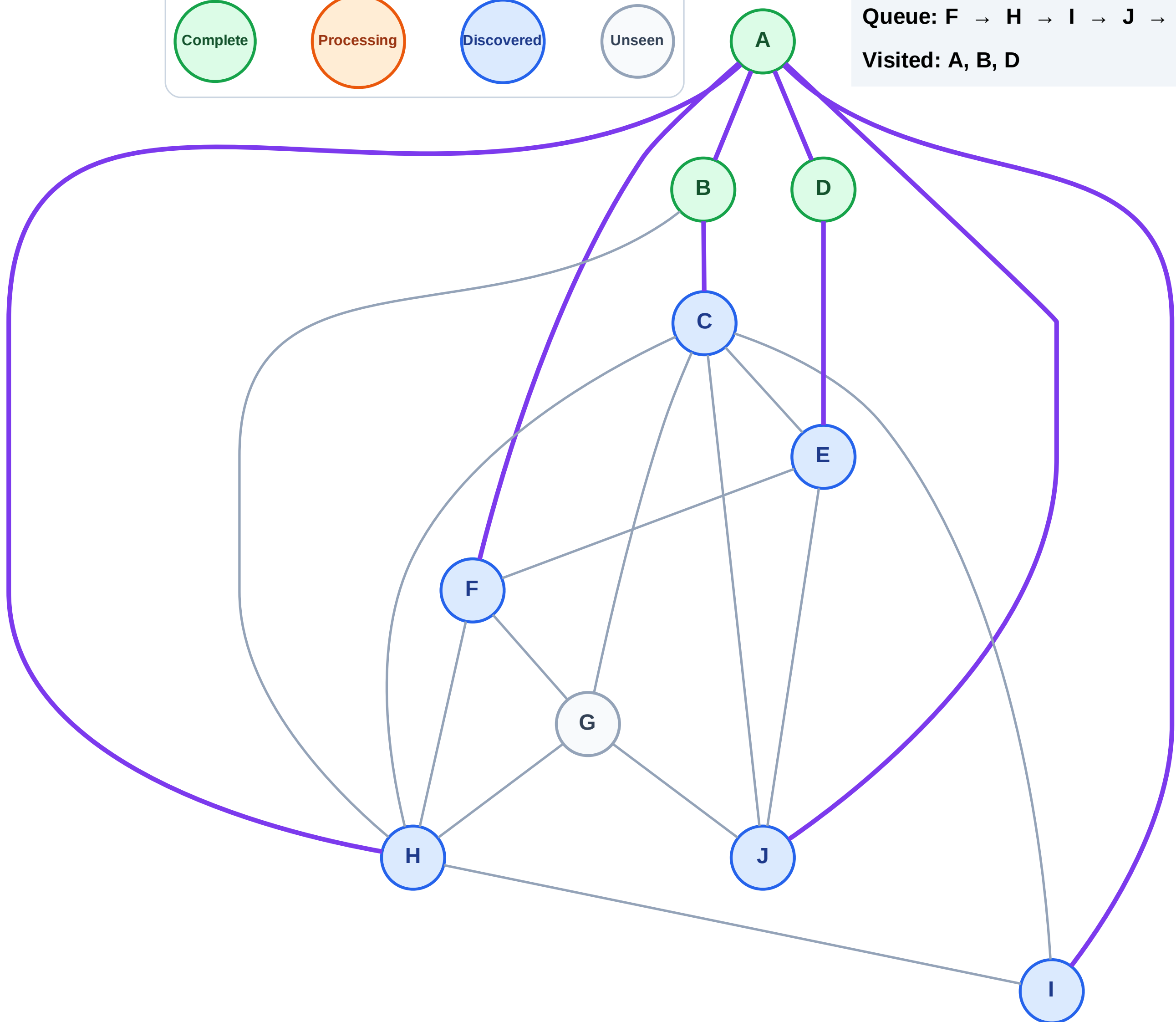
Step 7: Discover E from D

Legend



Queue: F → H → I → J → C → E

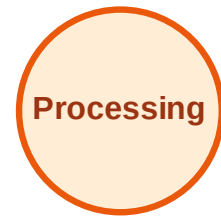
Visited: A, B, D



BFS Traversal

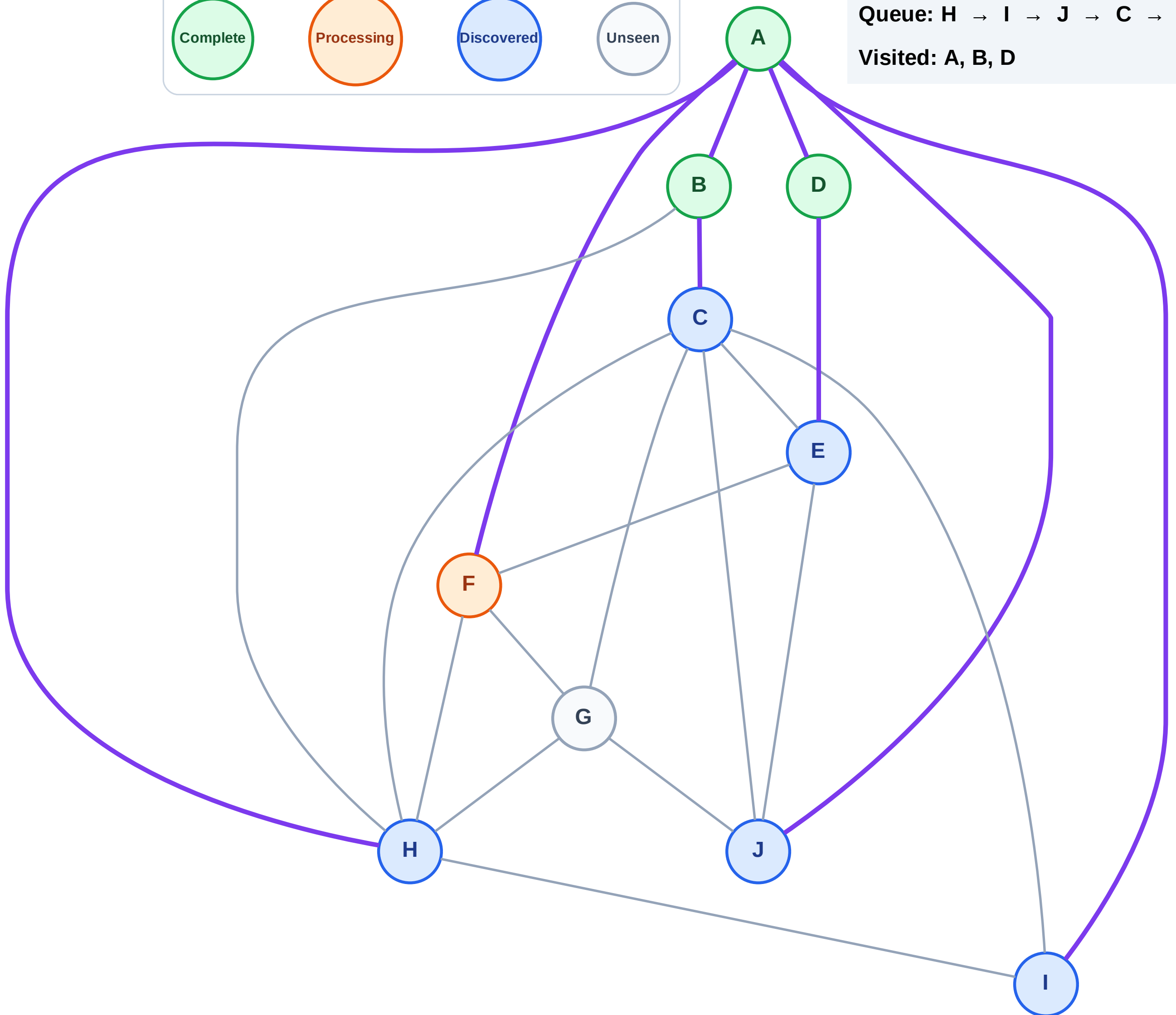
Step 8: Process node F

Legend



Queue: H → I → J → C → E

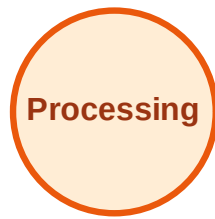
Visited: A, B, D



BFS Traversal

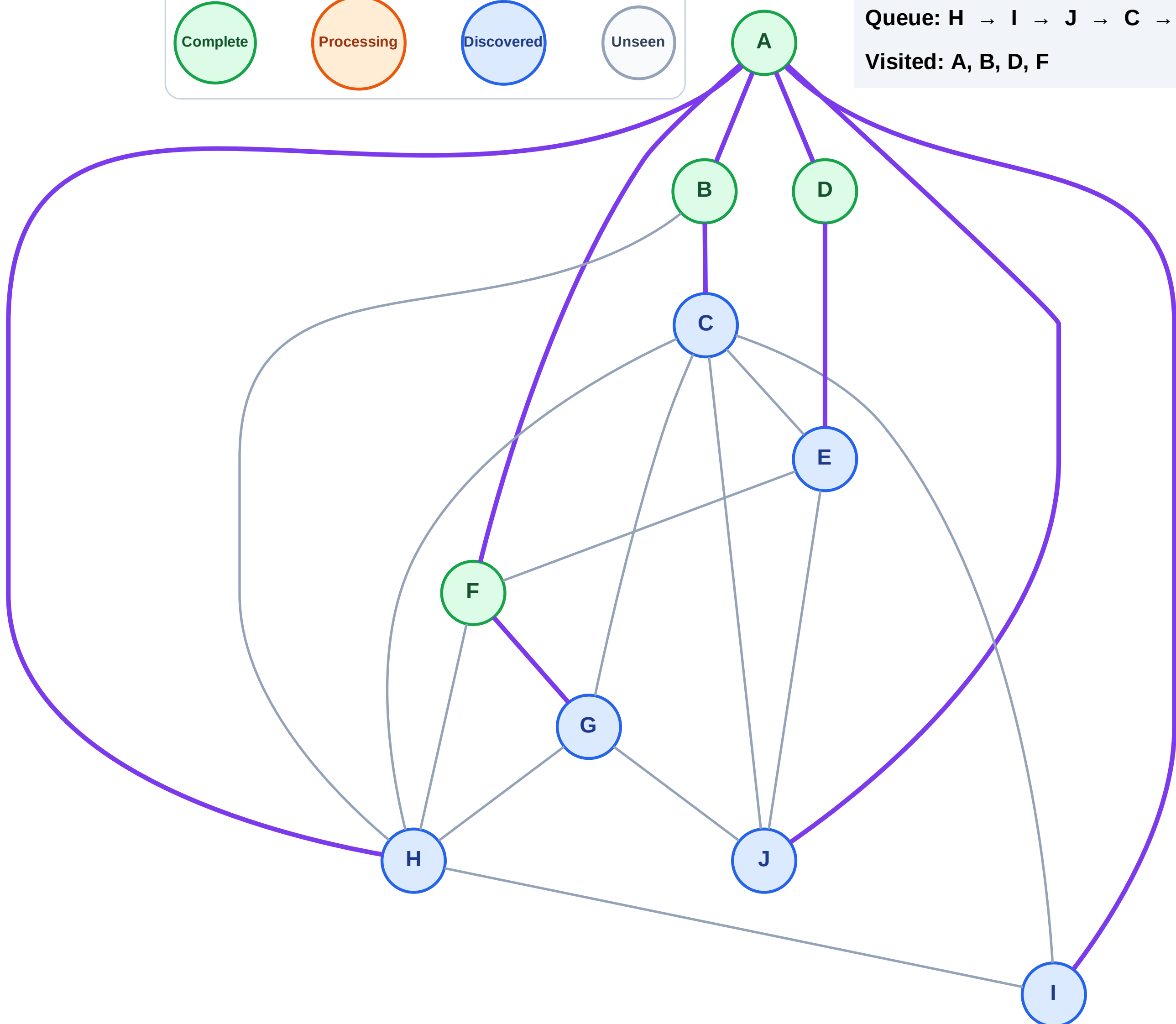
Step 9: Discover G from F

Legend



Queue: H → I → J → C → E → G

Visited: A, B, D, F



BFS Traversal

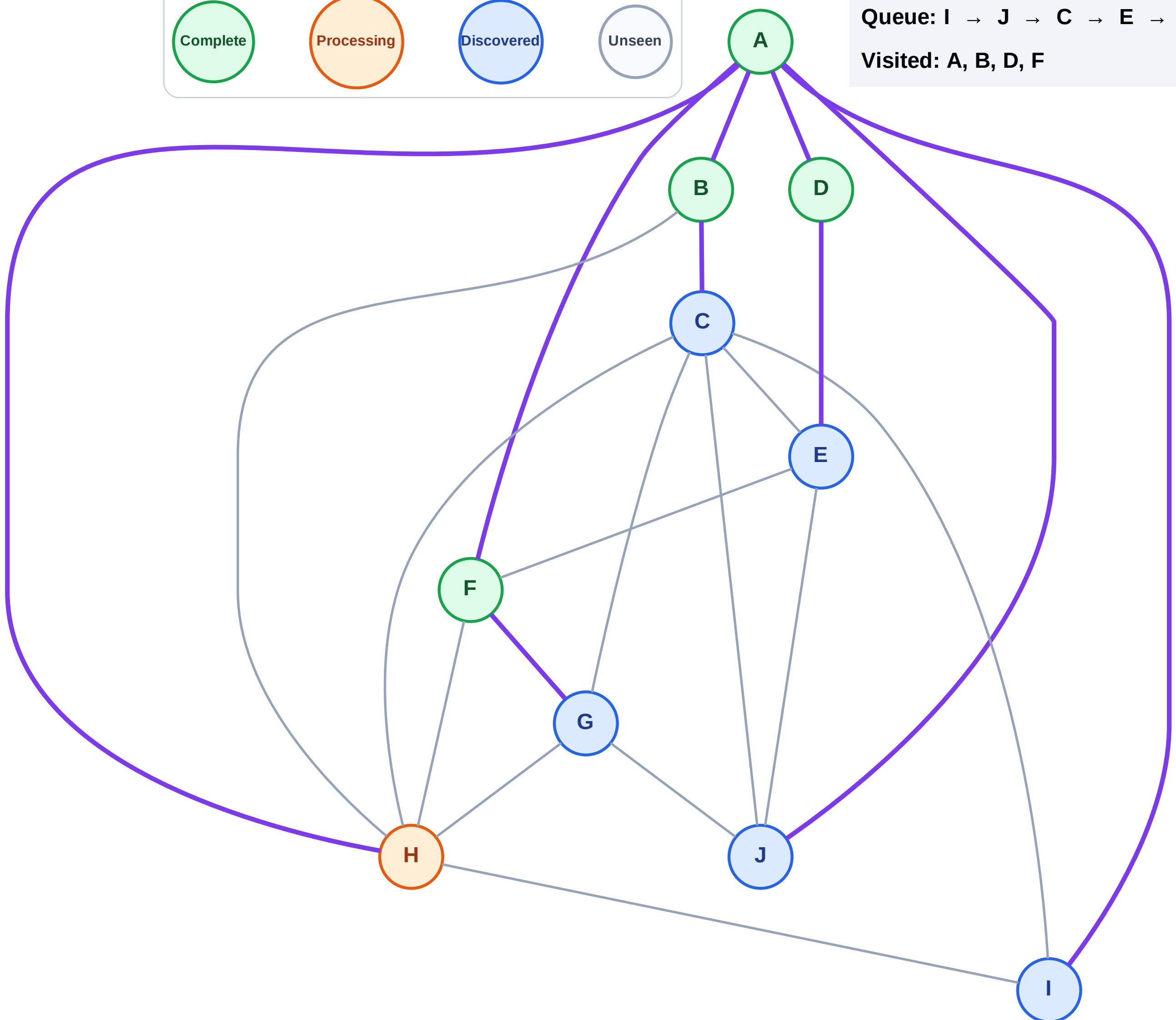
Step 10: Process node H

Legend



Queue: I → J → C → E → G

Visited: A, B, D, F



BFS Traversal

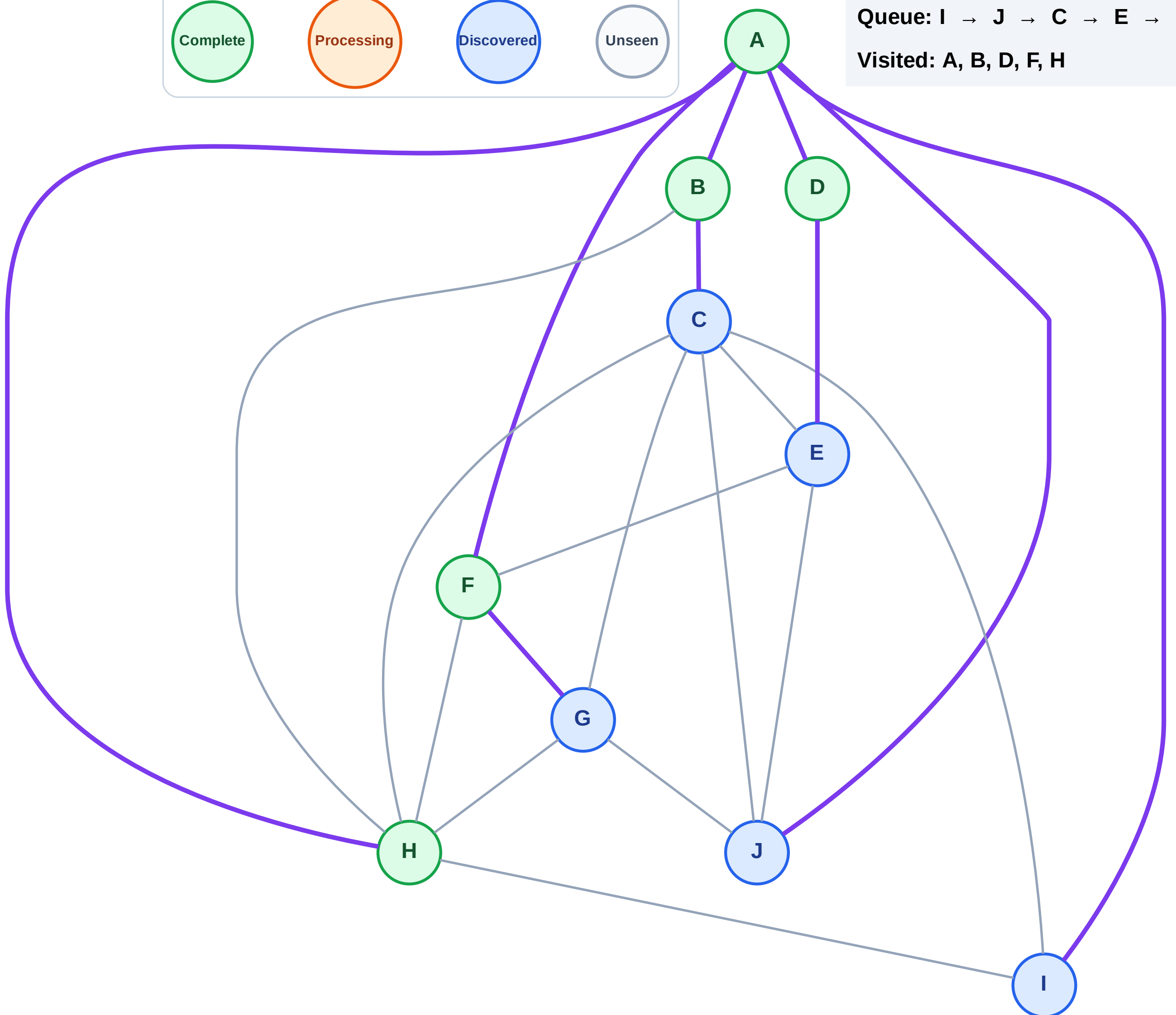
Step 11: No new neighbors from H

Legend



Queue: I → J → C → E → G

Visited: A, B, D, F, H



BFS Traversal

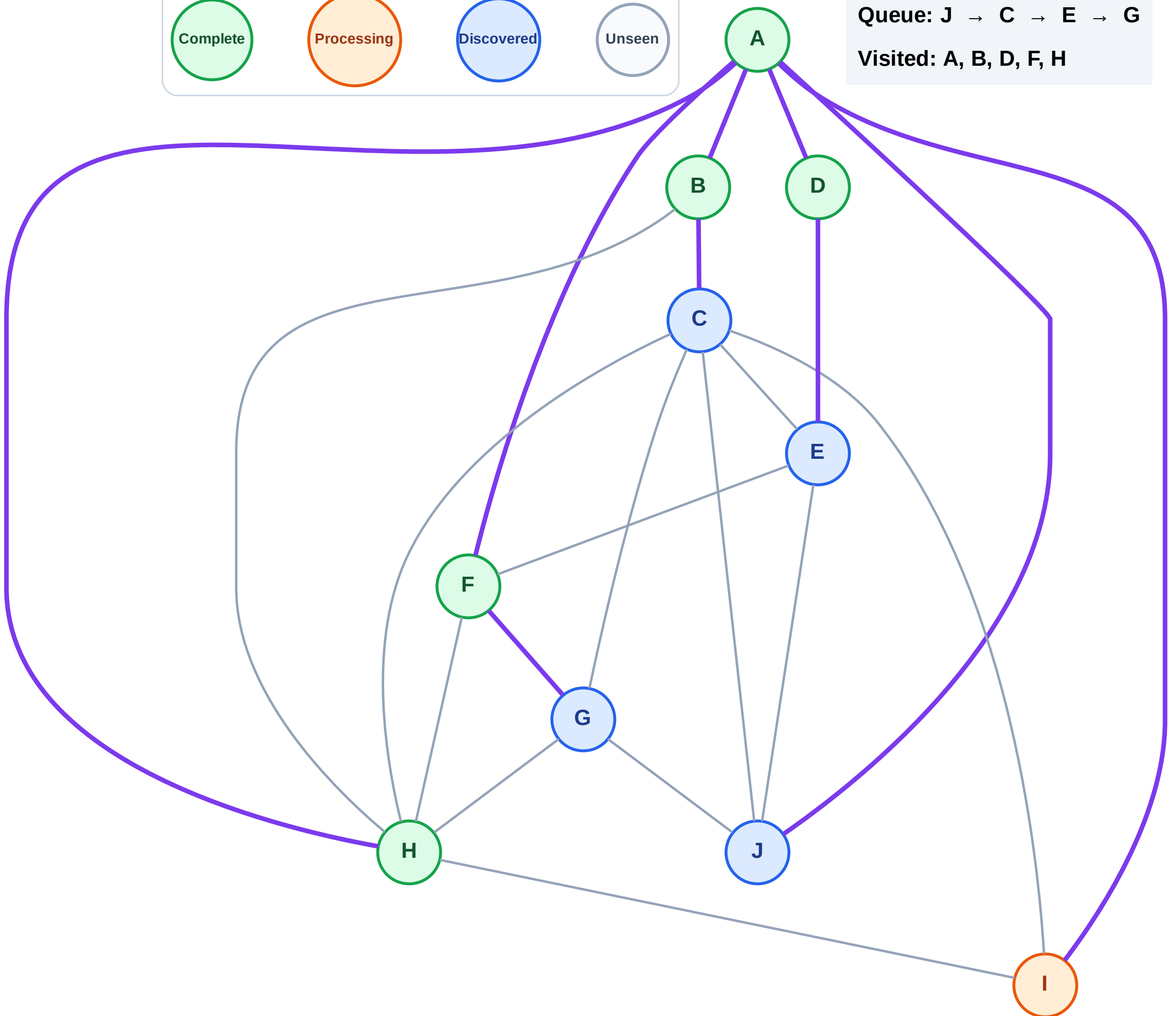
Step 12: Process node I

Legend



Queue: J → C → E → G

Visited: A, B, D, F, H



BFS Traversal

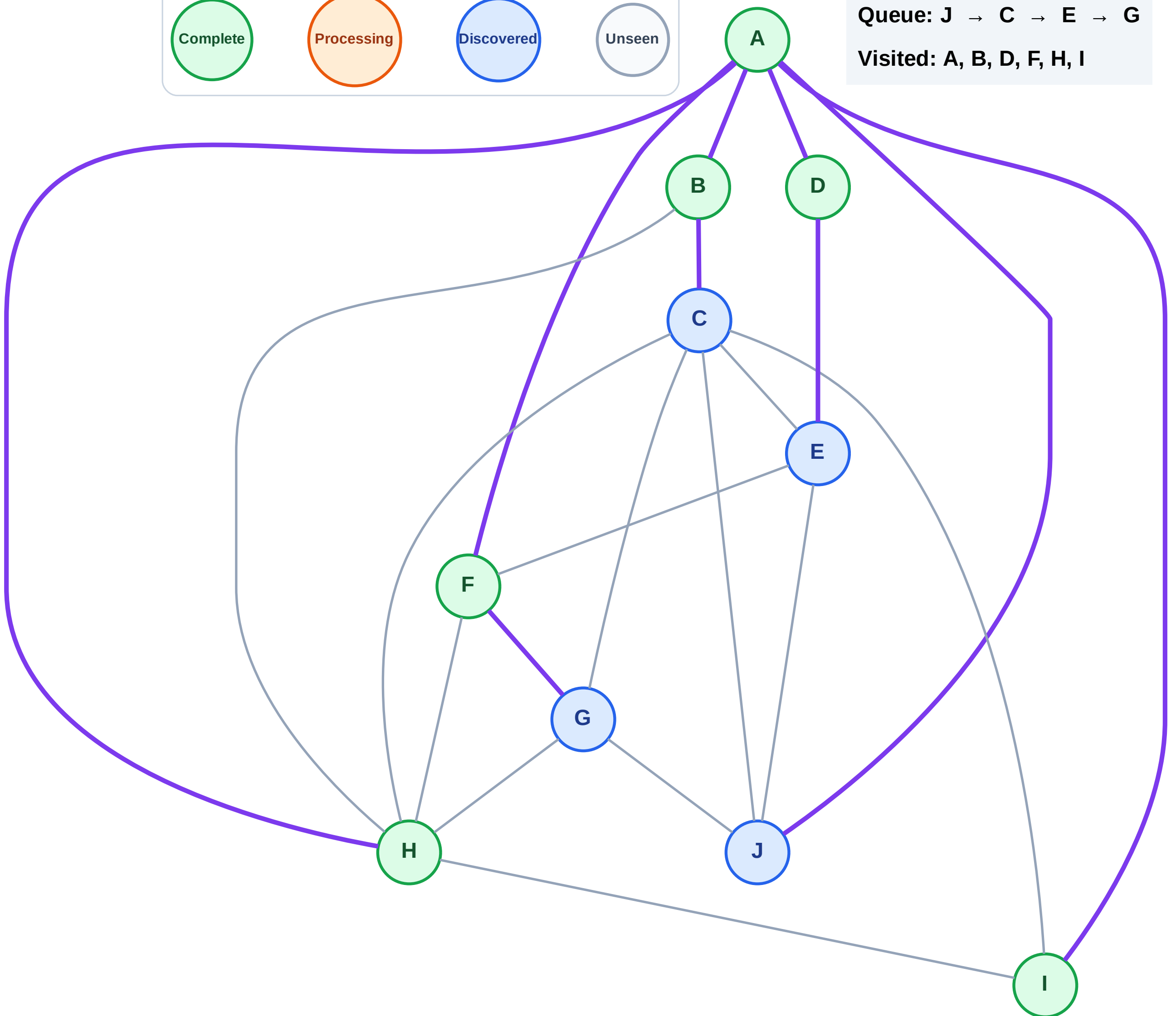
Step 13: No new neighbors from I

Legend



Queue: J → C → E → G

Visited: A, B, D, F, H, I



BFS Traversal

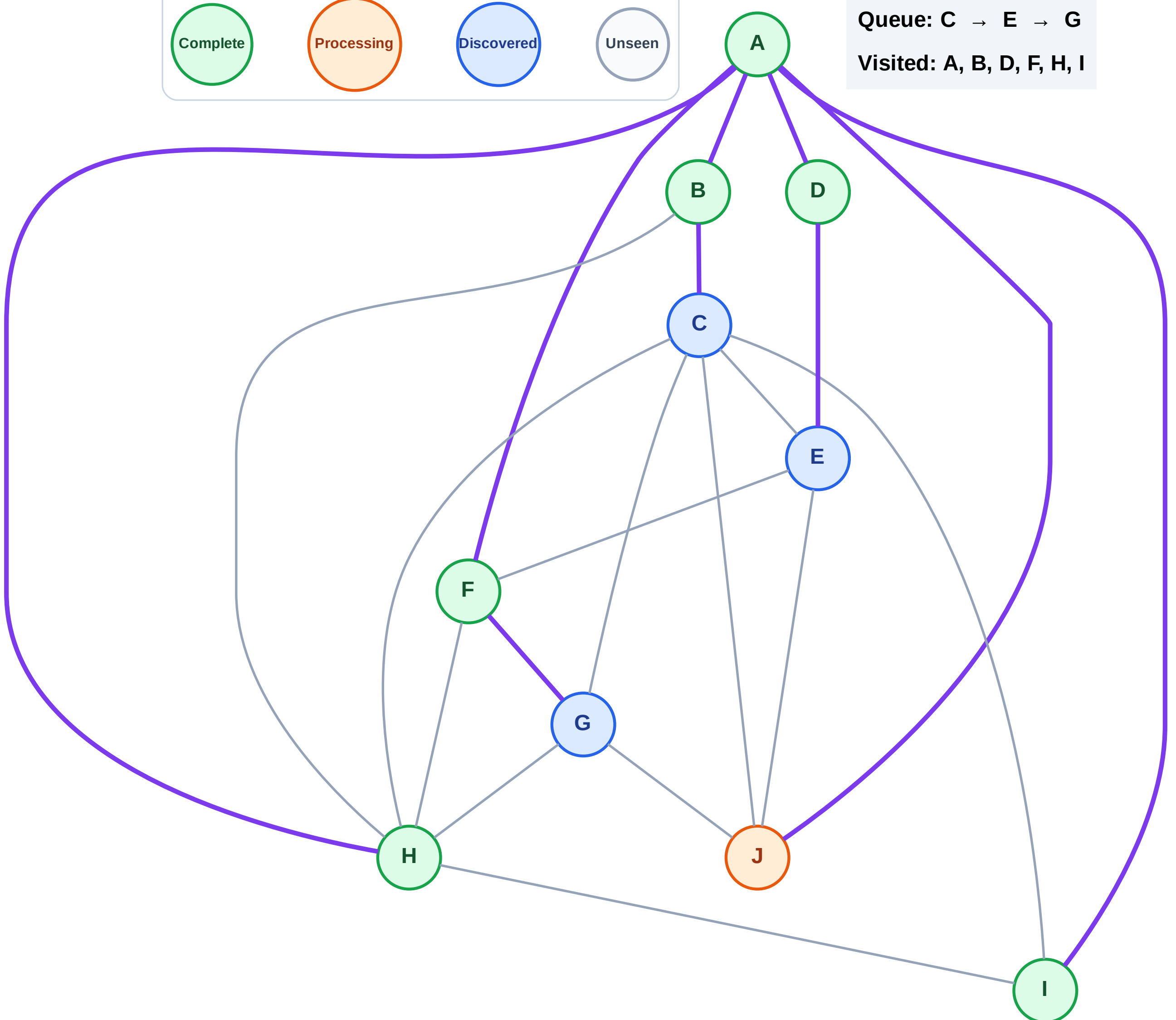
Step 14: Process node J

Legend



Queue: C → E → G

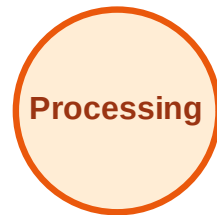
Visited: A, B, D, F, H, I



BFS Traversal

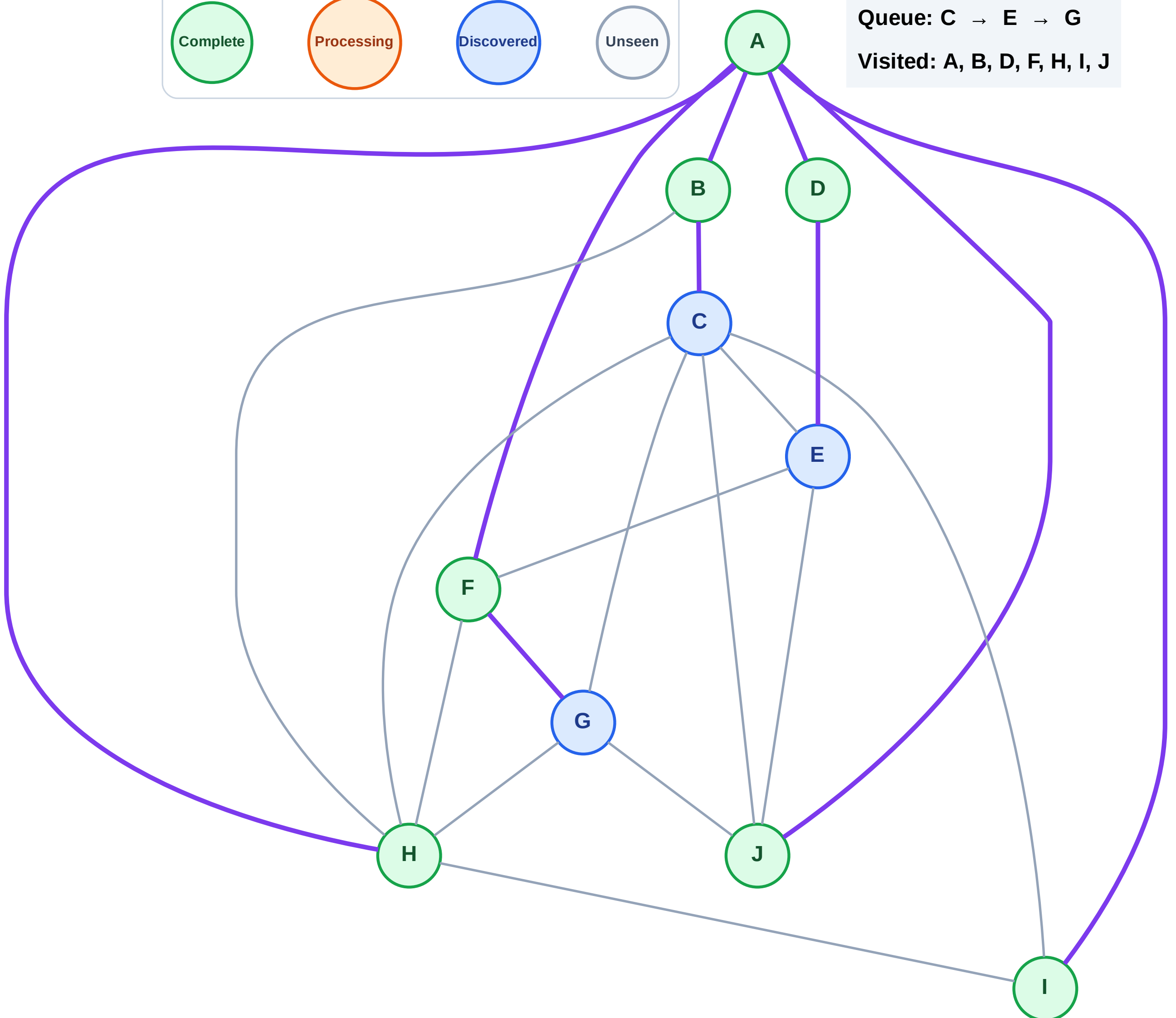
Step 15: No new neighbors from J

Legend



Queue: C → E → G

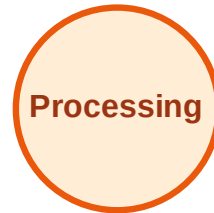
Visited: A, B, D, F, H, I, J



BFS Traversal

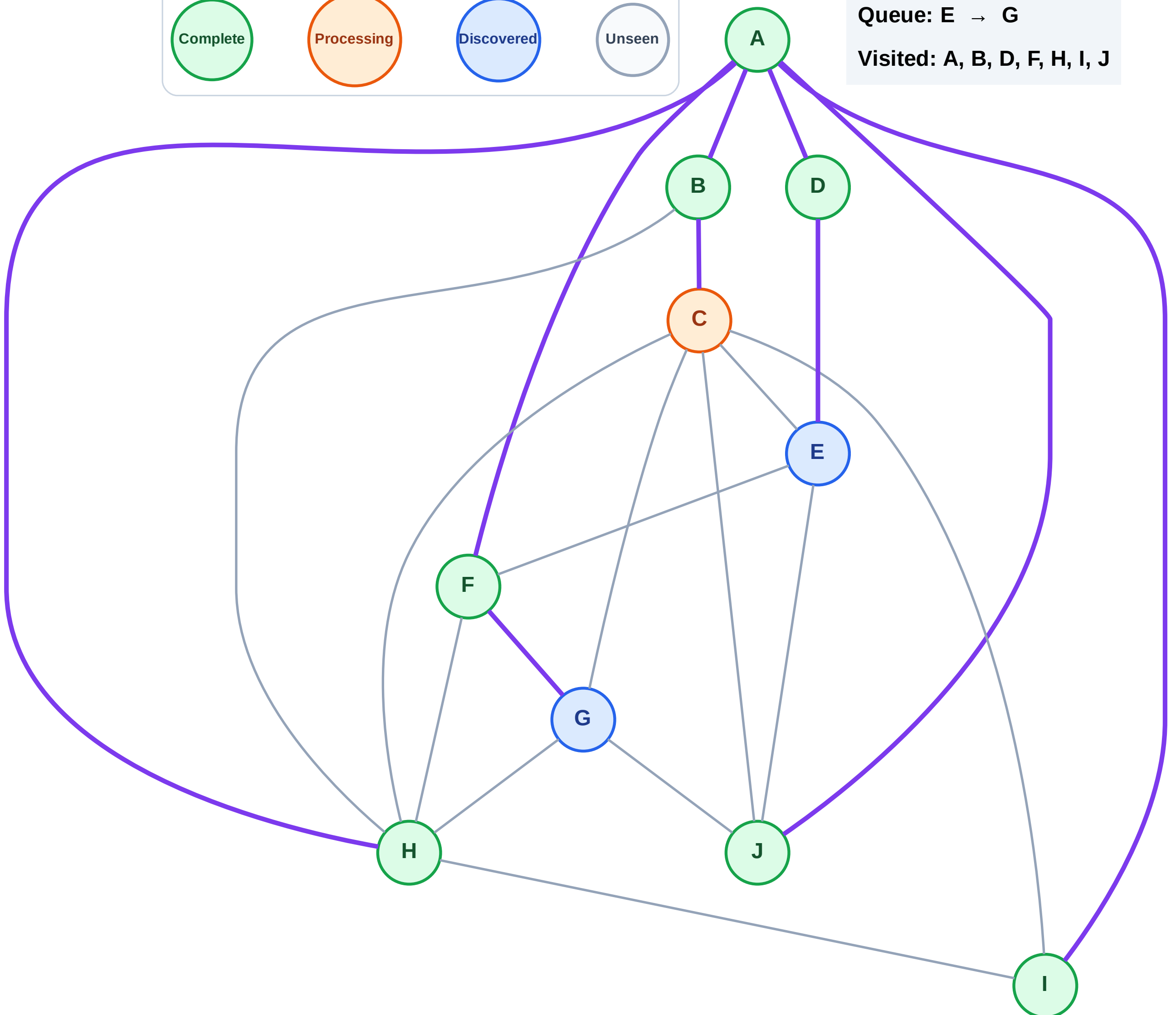
Step 16: Process node C

Legend



Queue: E → G

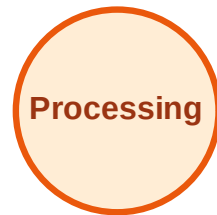
Visited: A, B, D, F, H, I, J



BFS Traversal

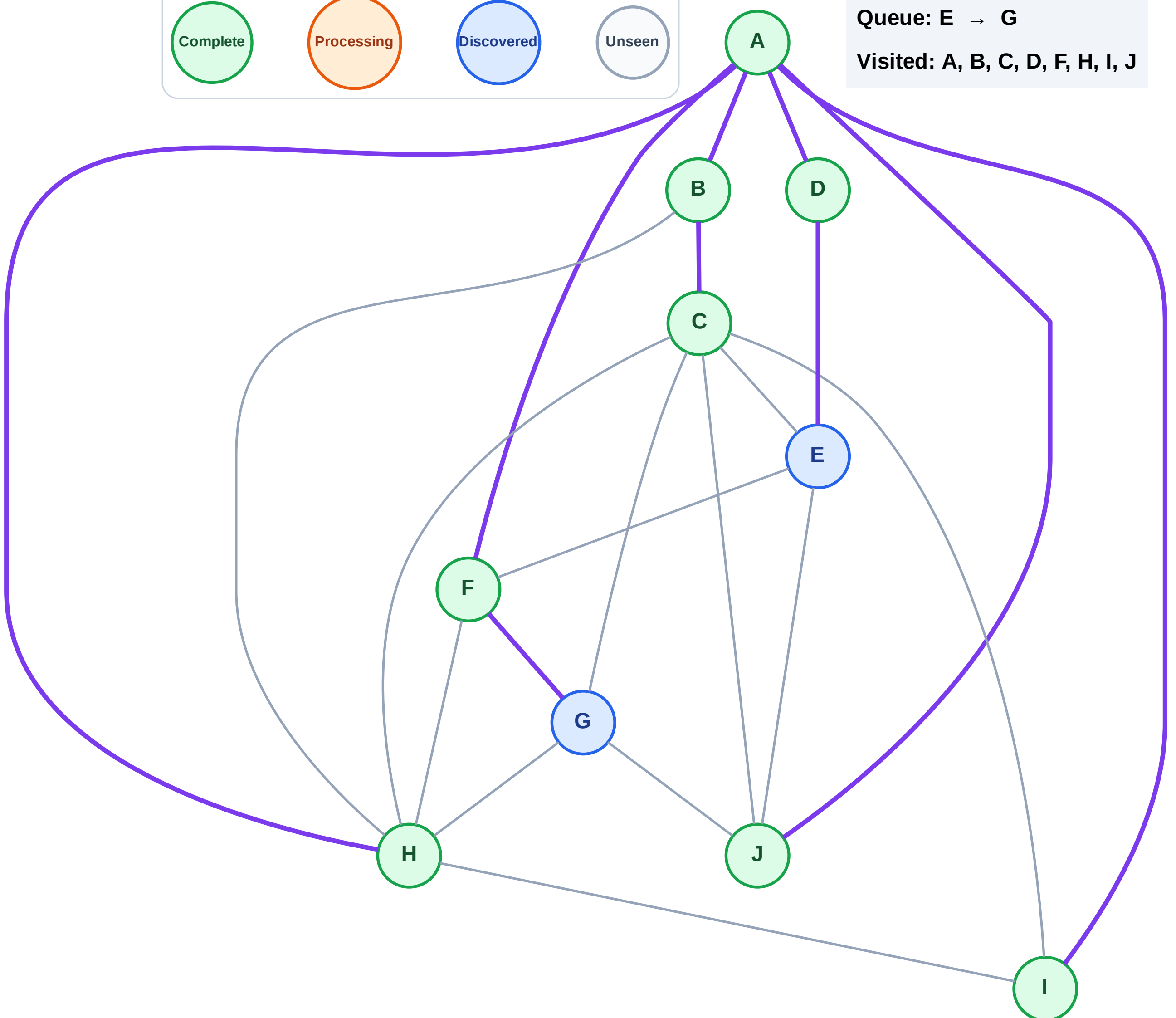
Step 17: No new neighbors from C

Legend



Queue: E → G

Visited: A, B, C, D, F, H, I, J

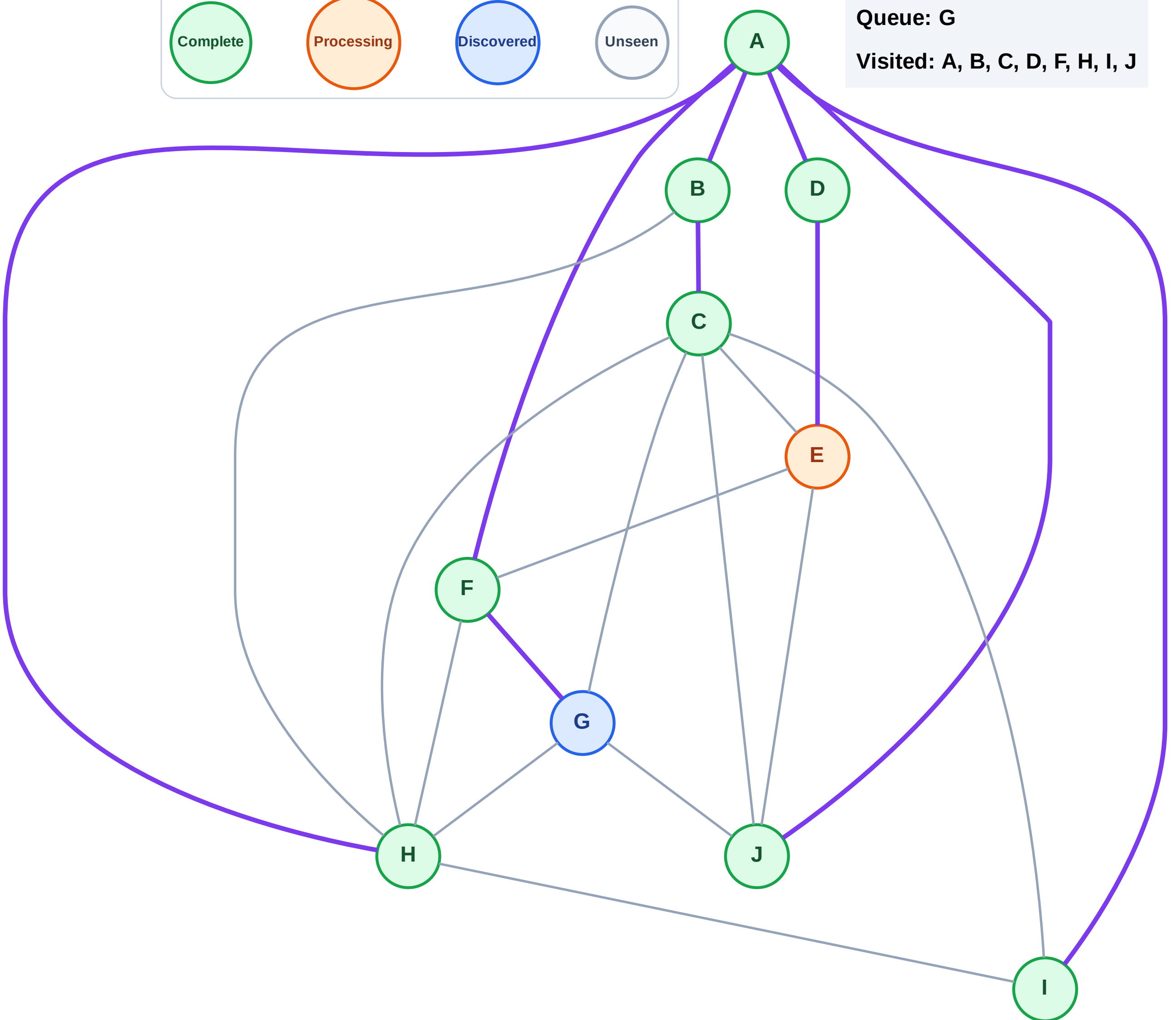


Step 18: Process node E

Legend

Unseen

Visited: A, B, C, D, F, H, I, J



BFS Traversal

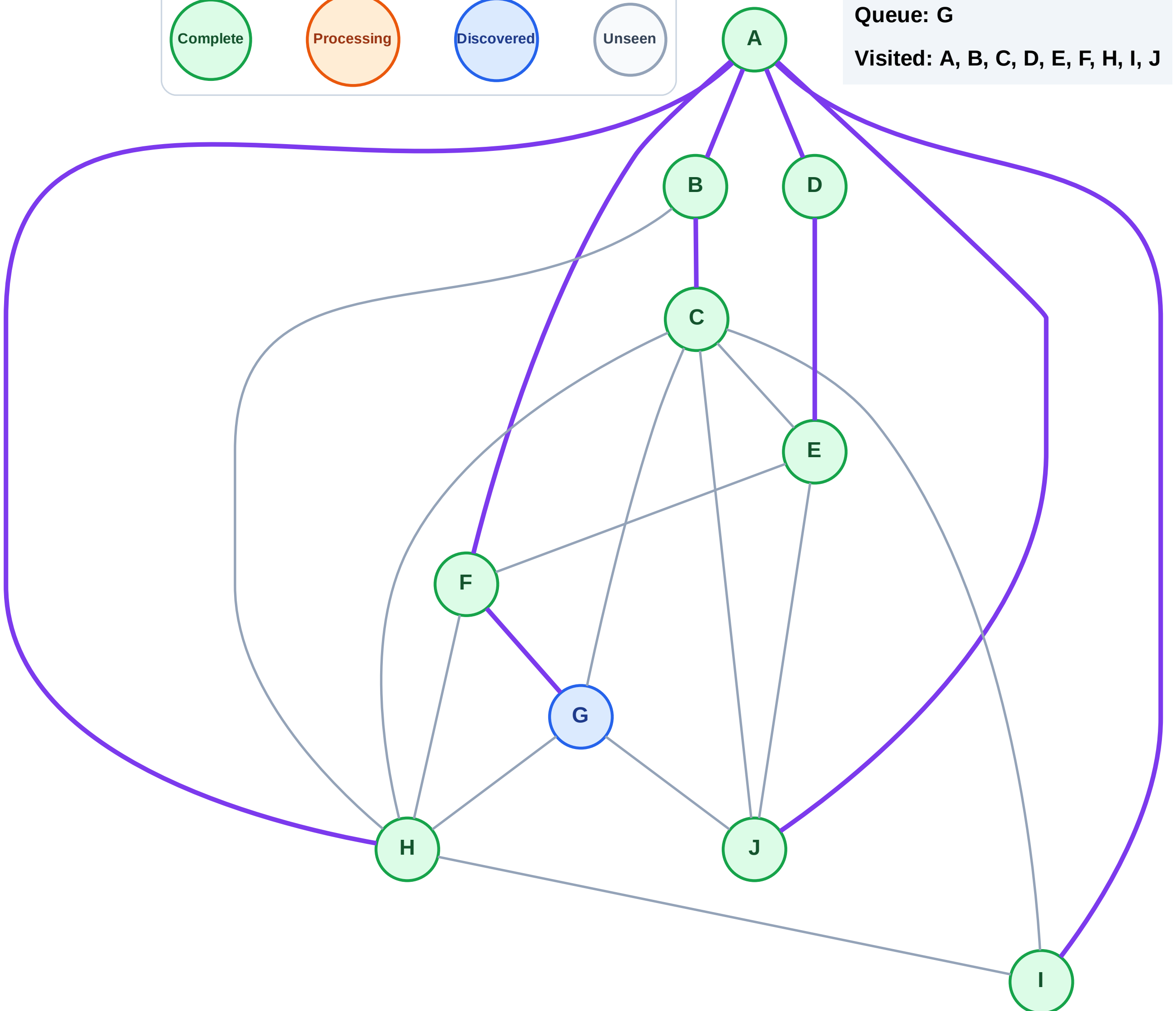
Step 19: No new neighbors from E

Legend



Queue: G

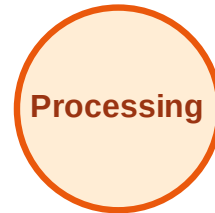
Visited: A, B, C, D, E, F, H, I, J



BFS Traversal

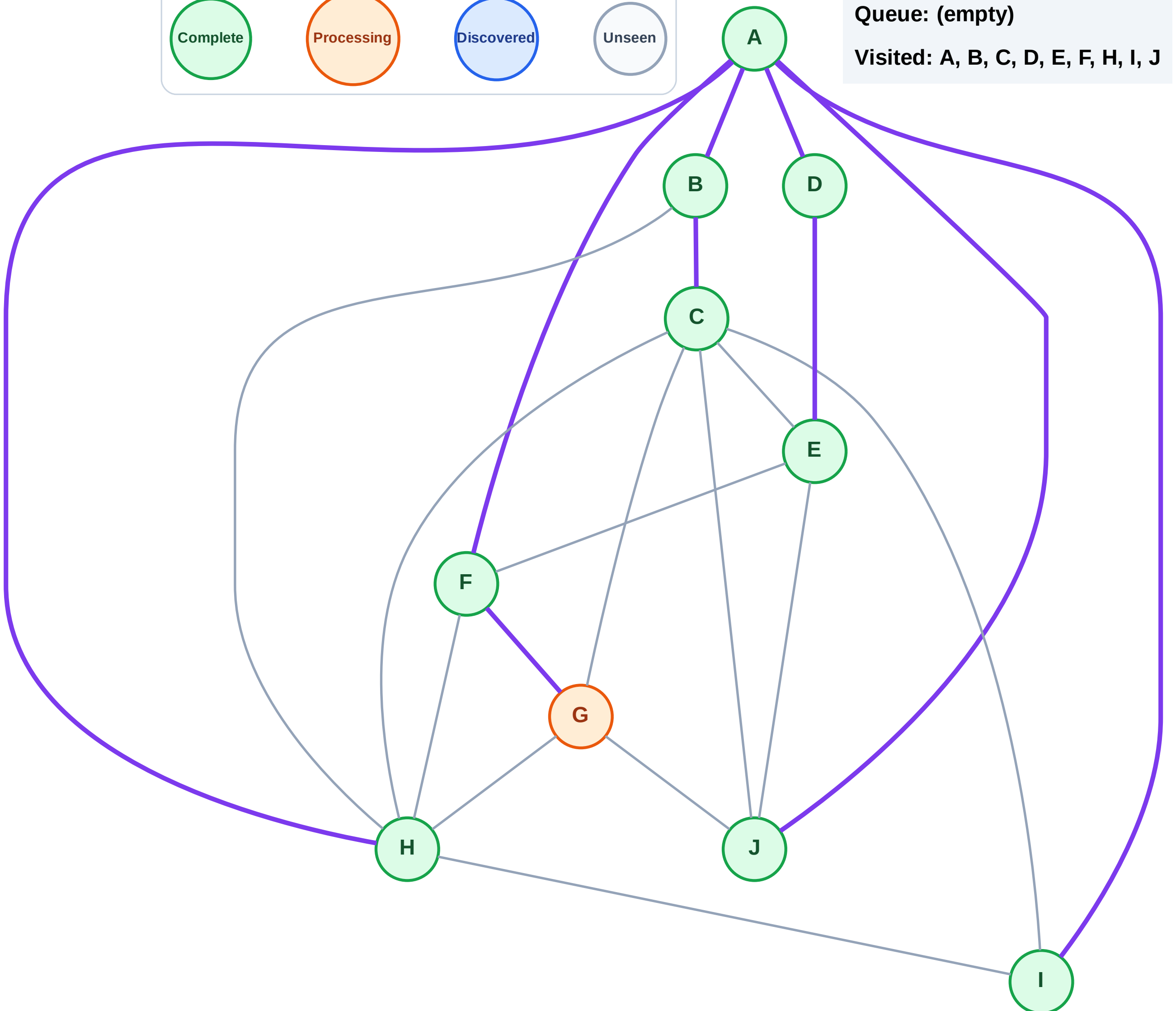
Step 20: Process node G

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, H, I, J



BFS Traversal

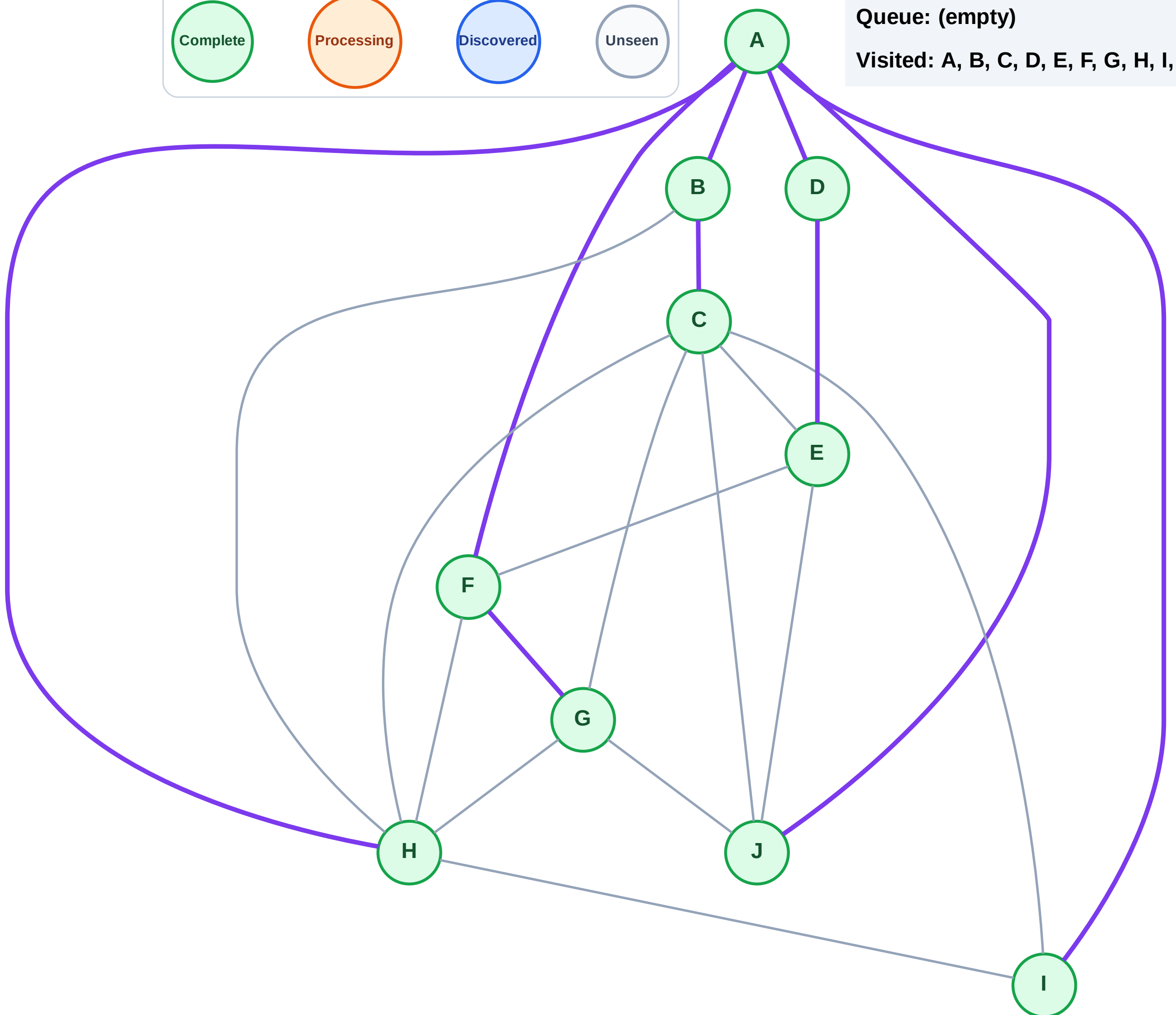
Step 21: No new neighbors from G

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J



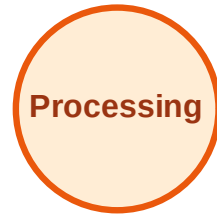
DFS Traversal

Step 1: Start at A

Legend



Complete



Processing



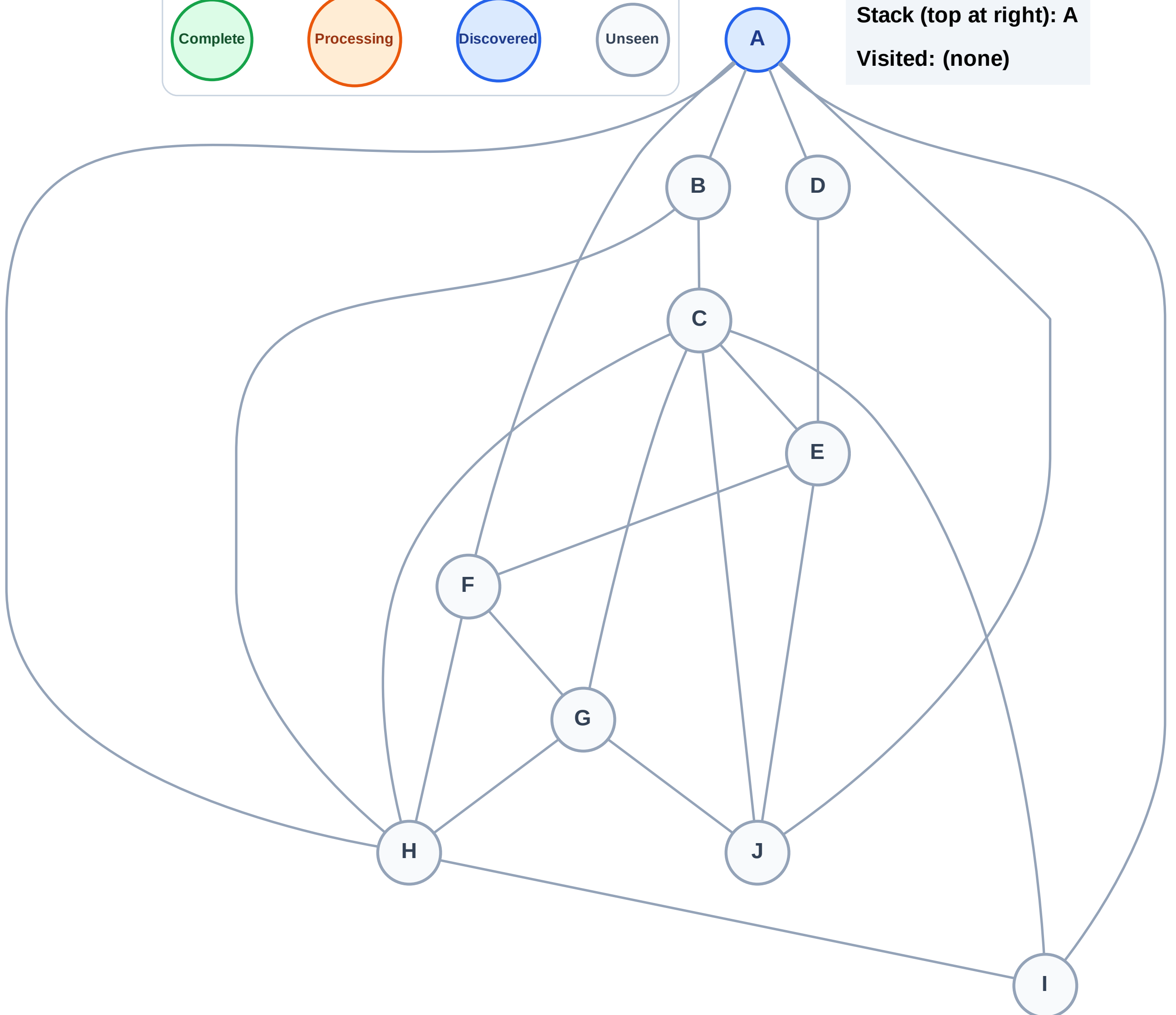
Discovered



Unseen

Stack (top at right): A

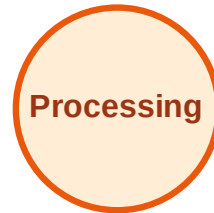
Visited: (none)



DFS Traversal

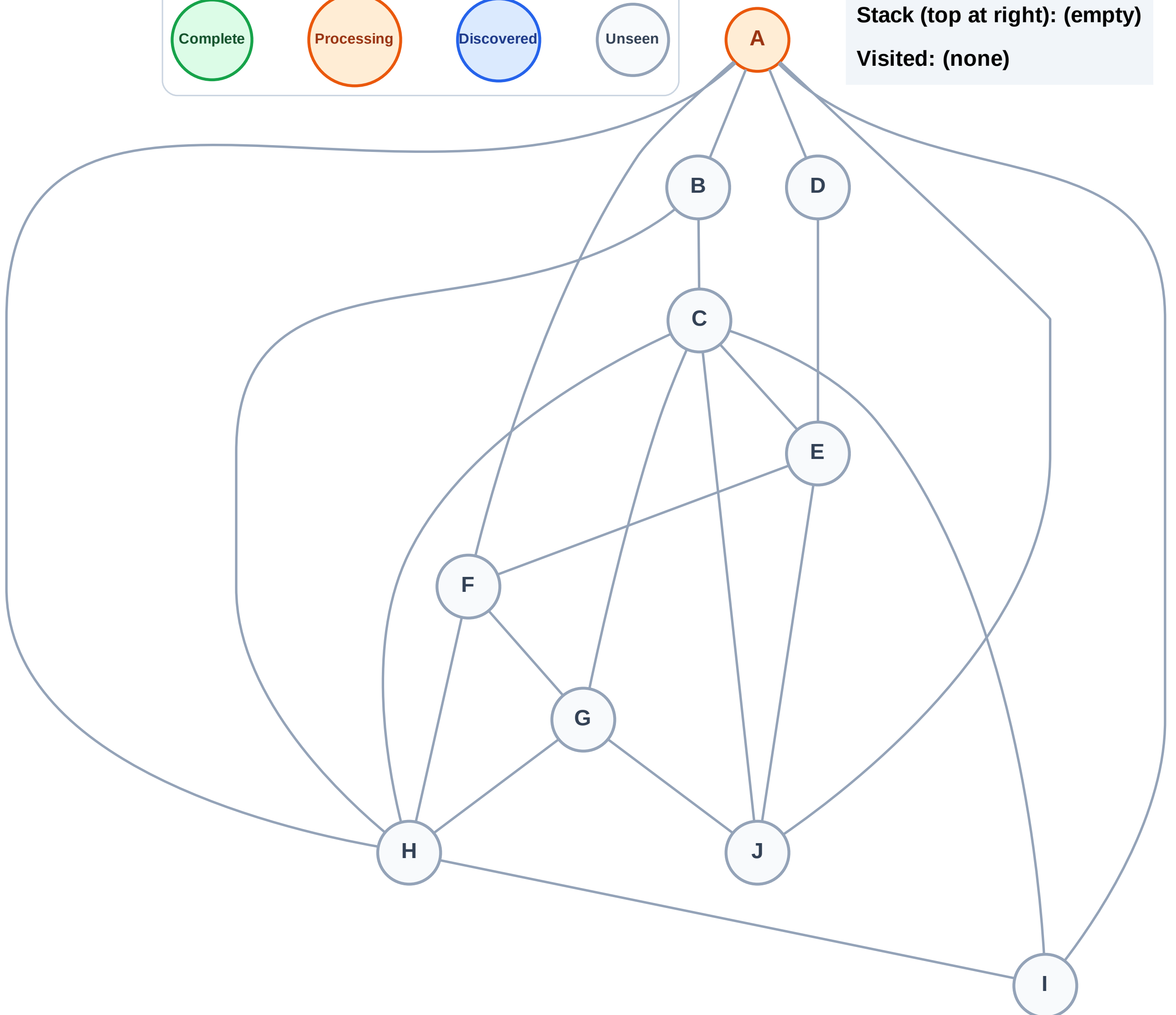
Step 2: Process node A

Legend



Stack (top at right): (empty)

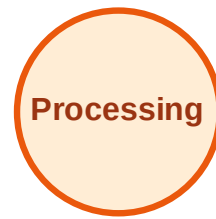
Visited: (none)



DFS Traversal

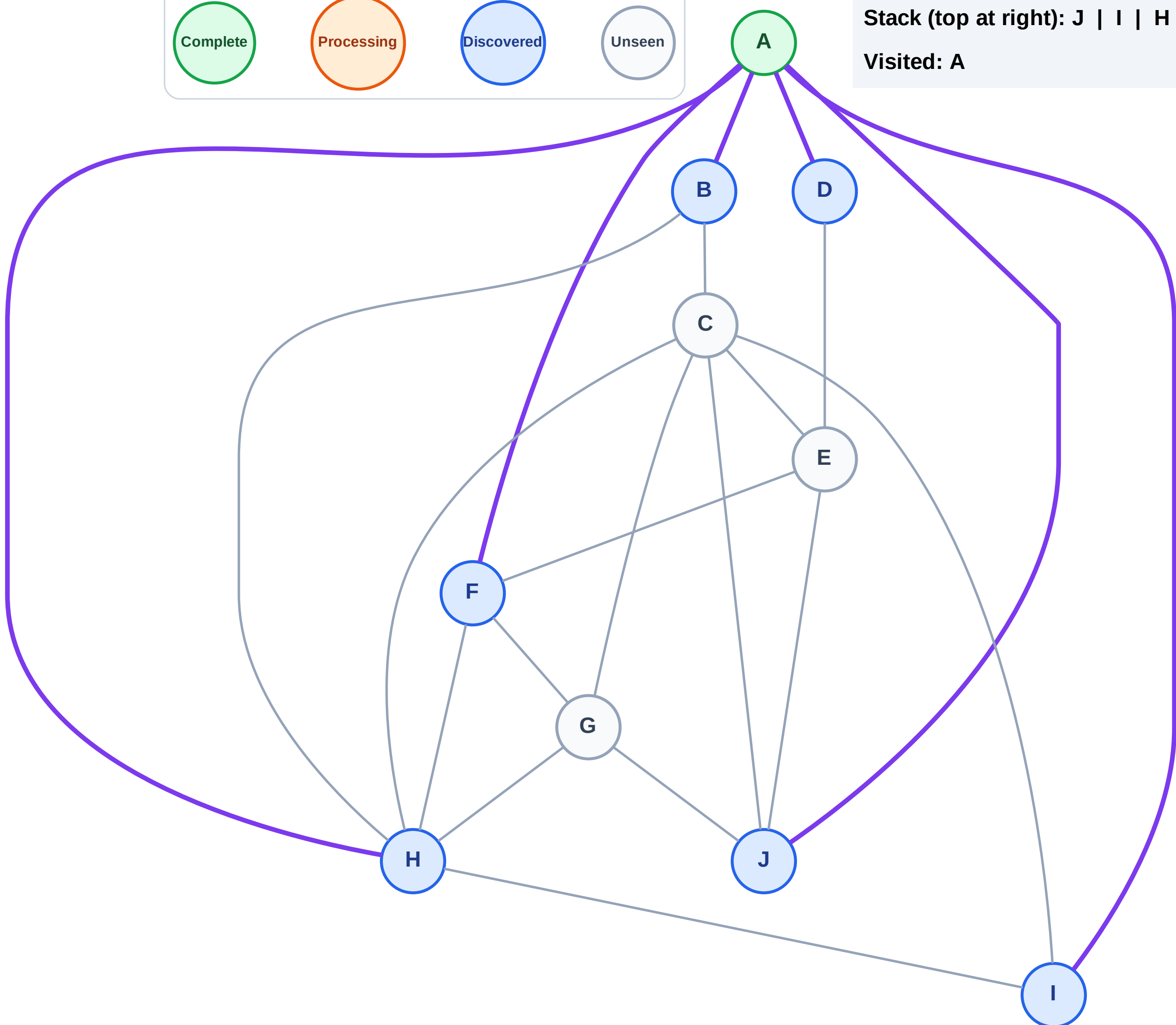
Step 3: Discover J, I, H, F, D, B from A

Legend



Stack (top at right): J | I | H | F | D | B

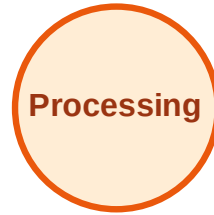
Visited: A



DFS Traversal

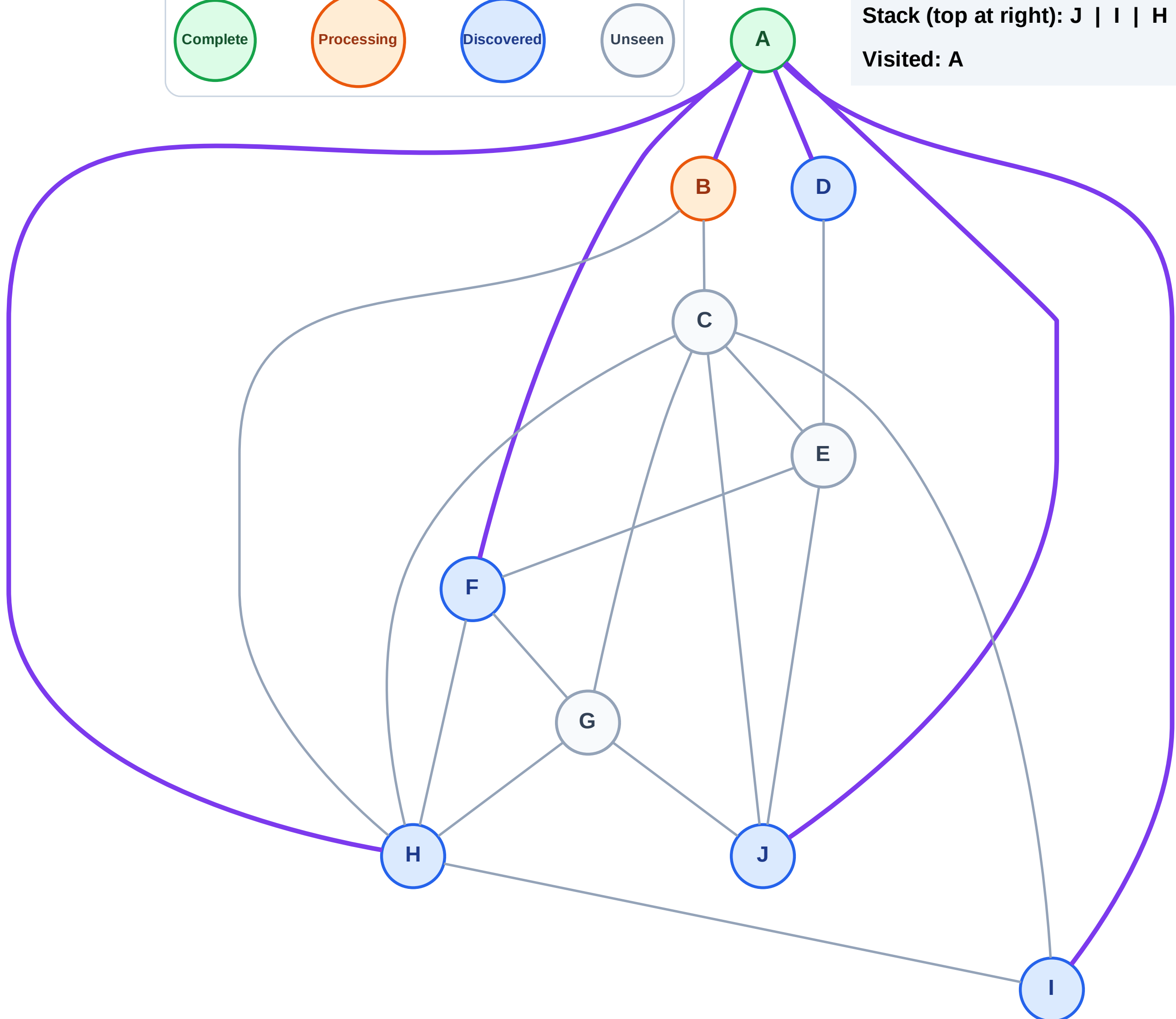
Step 4: Process node B

Legend



Stack (top at right): J | I | H | F | D

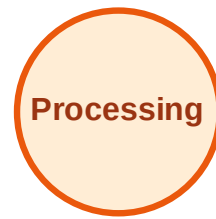
Visited: A



DFS Traversal

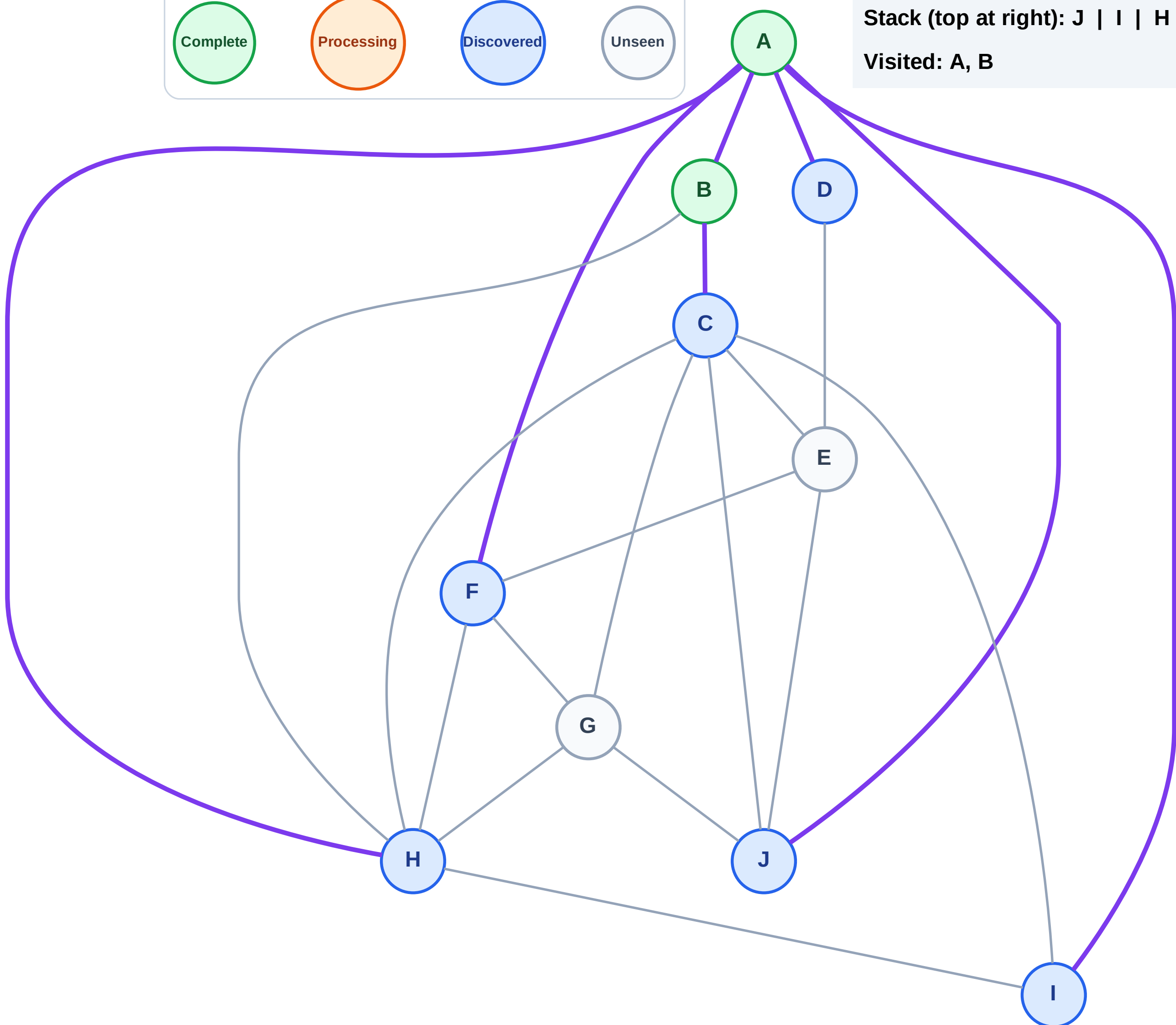
Step 5: Discover C from B

Legend



Stack (top at right): J | I | H | F | D | C

Visited: A, B



Step 6: Process node C

Step 6: Process node C

Legend



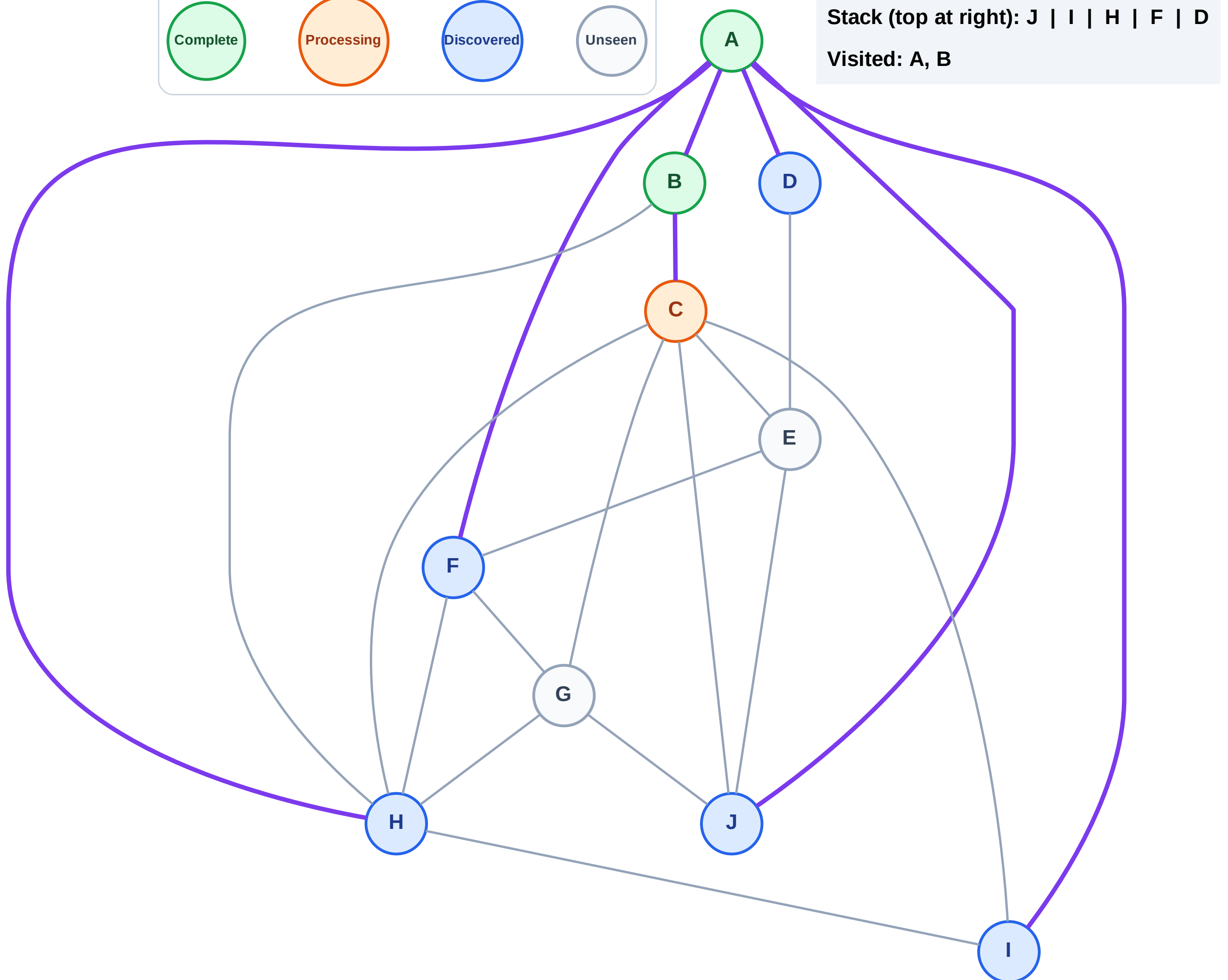
The legend consists of four colored circles arranged horizontally, each containing a label. From left to right: a green circle labeled 'Complete', an orange circle labeled 'Processing', a blue circle labeled 'Discovered', and a grey circle labeled 'Unseen'.

Processing

Unseen

Stack (top at right): J | I | H | F | D

Visited: A, B



DFS Traversal

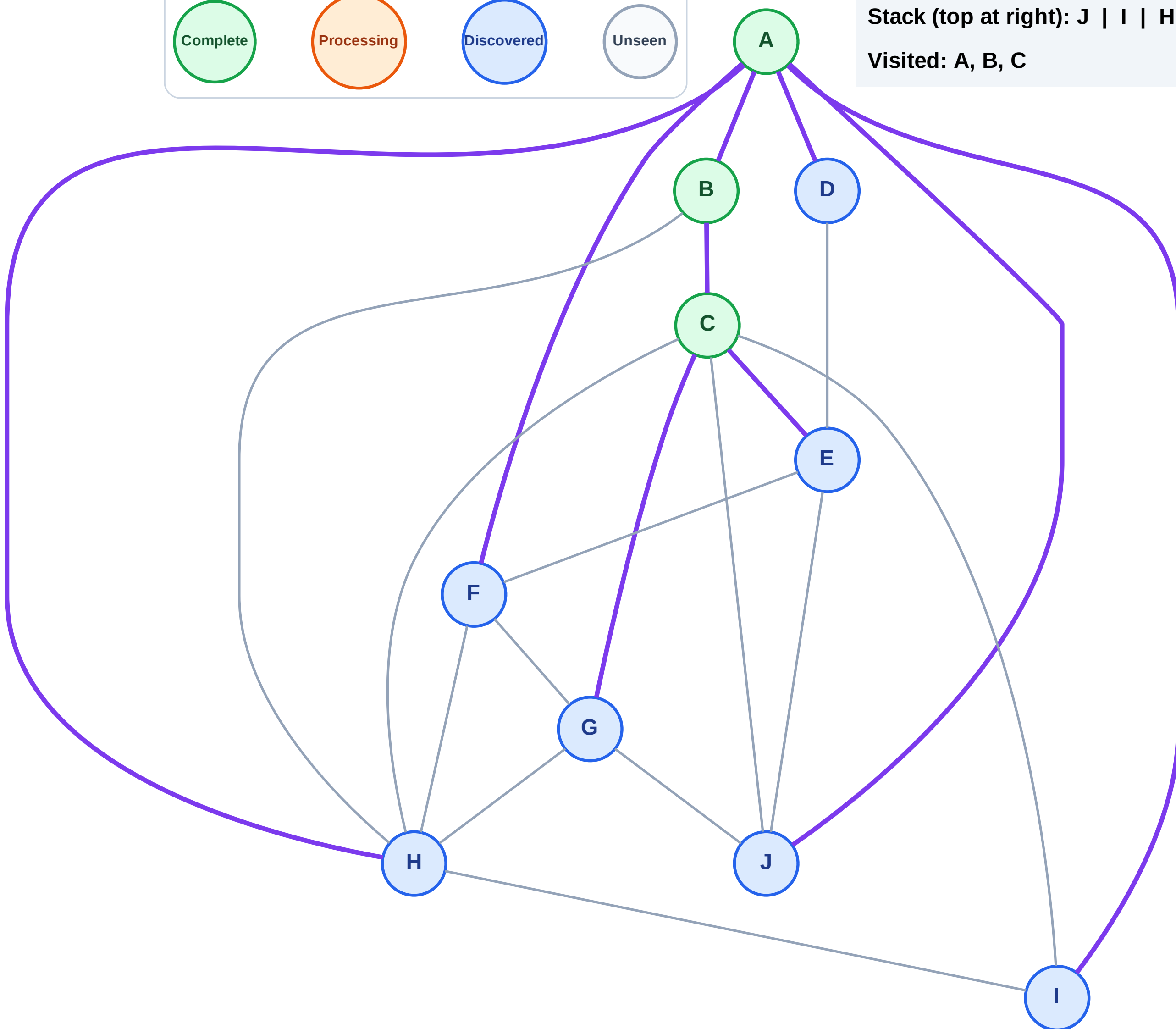
Step 7: Discover G, E from C

Legend



Stack (top at right): J | I | H | F | D | G | E

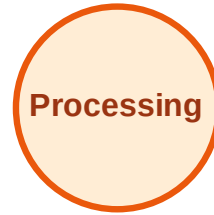
Visited: A, B, C



DFS Traversal

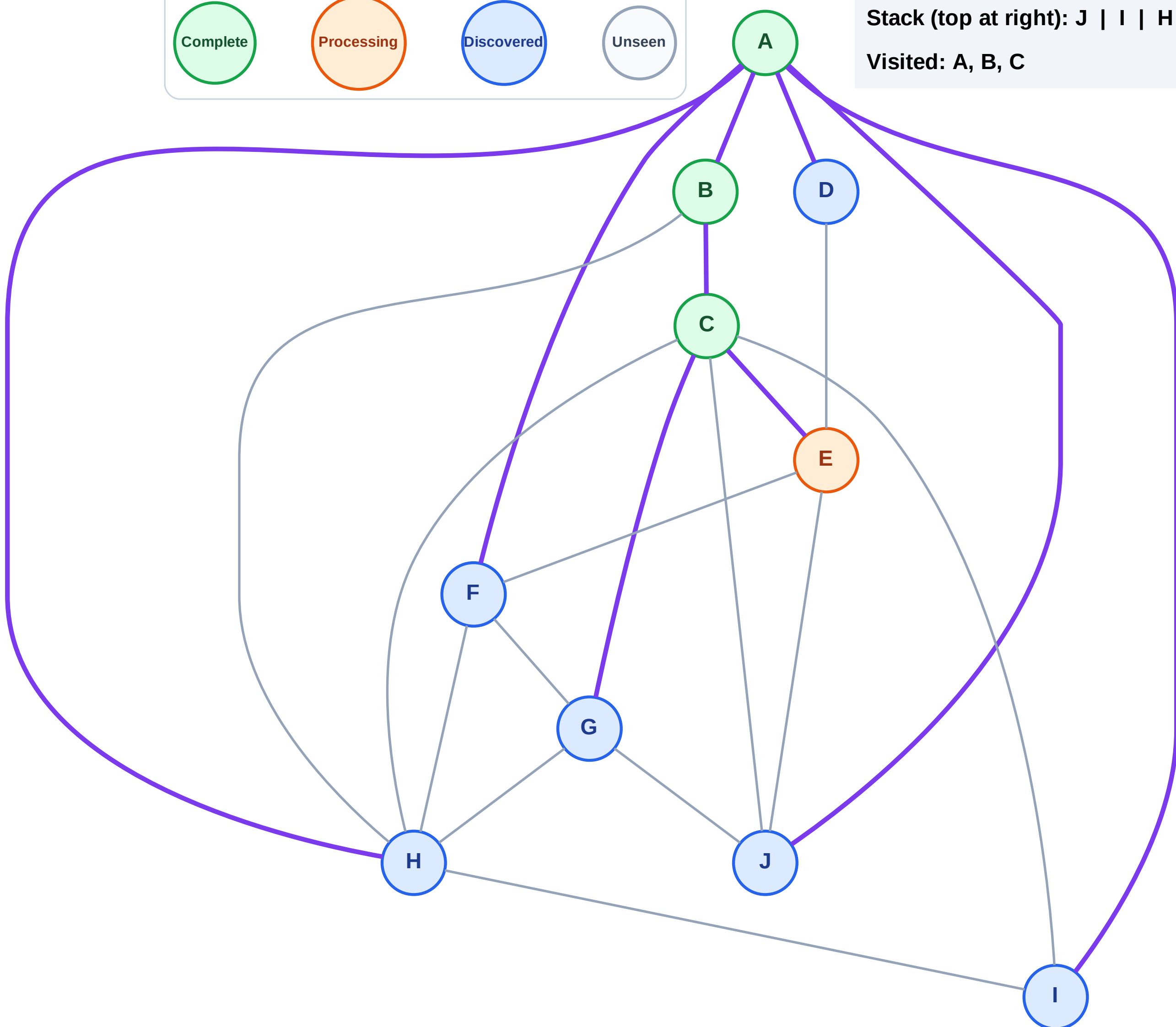
Step 8: Process node E

Legend



Stack (top at right): J | I | H | F | D | G

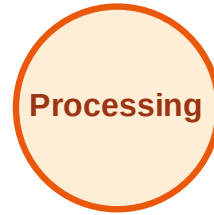
Visited: A, B, C



DFS Traversal

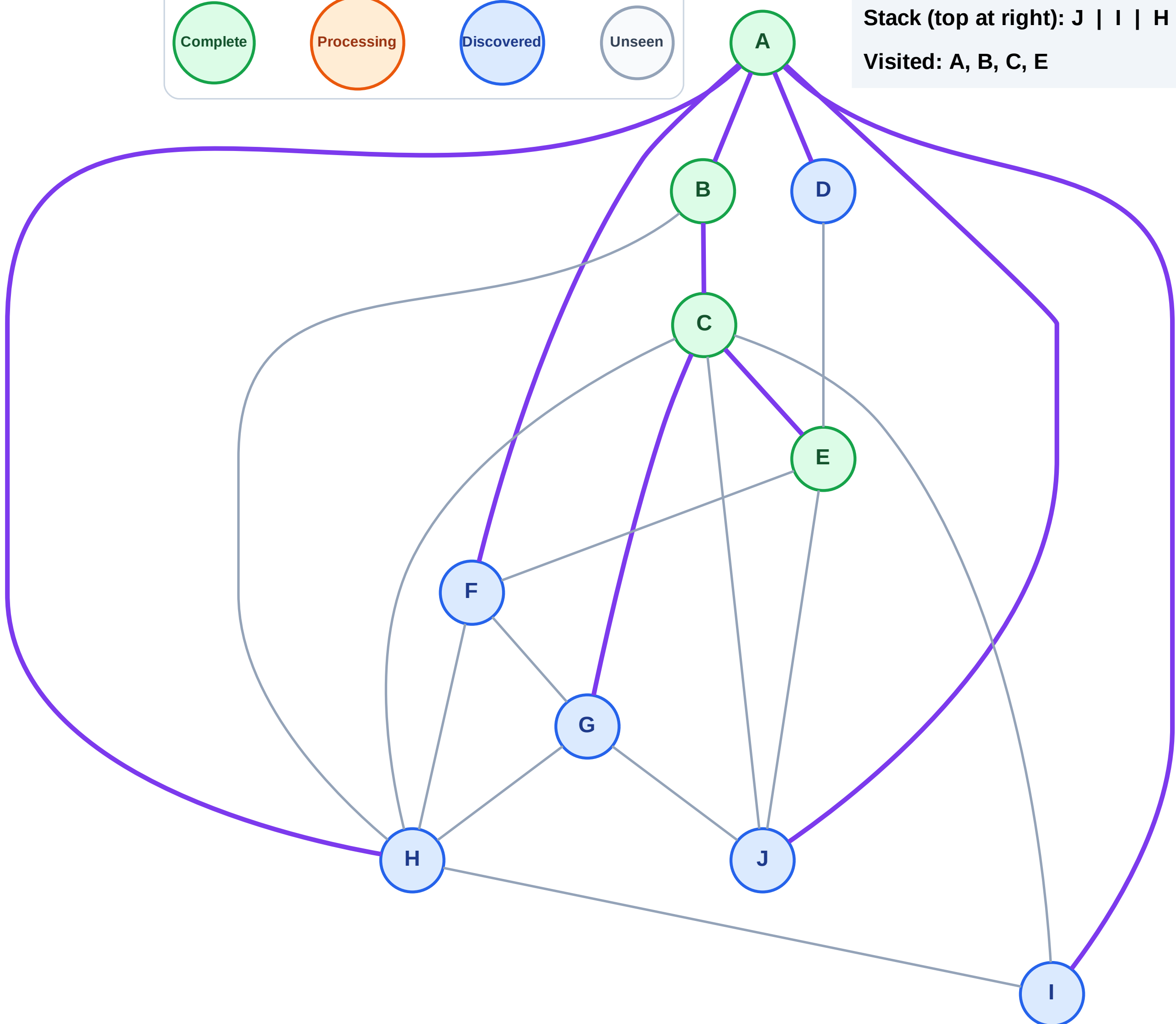
Step 9: No new neighbors from E

Legend



Stack (top at right): J | I | H | F | D | G

Visited: A, B, C, E



DFS Traversal

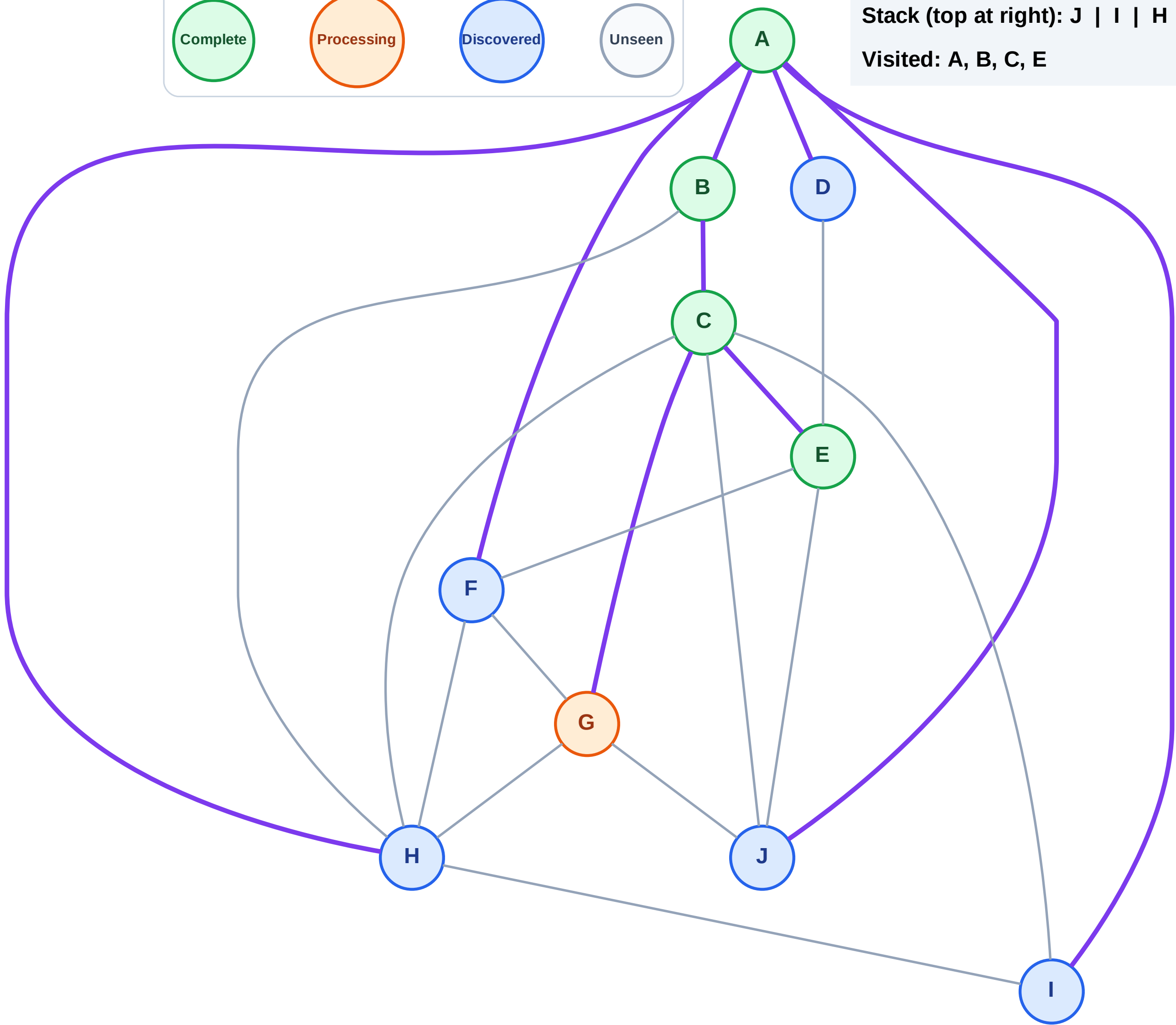
Step 10: Process node G

Legend



Stack (top at right): J | I | H | F | D

Visited: A, B, C, E



Step 11: No new neighbors from G

Step 11: No new neighbors from G



DFS Traversal

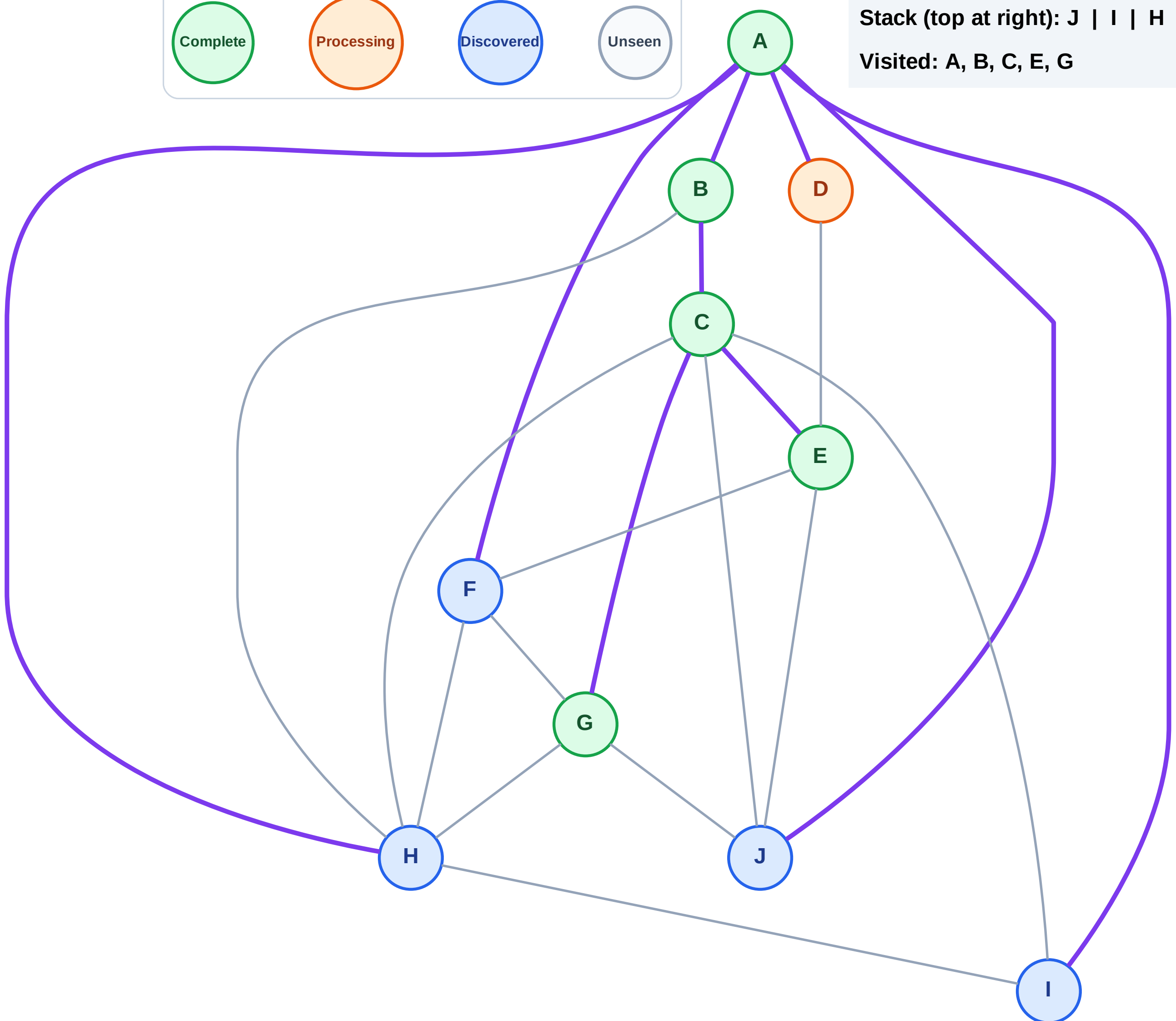
Step 12: Process node D

Legend



Stack (top at right): J | I | H | F

Visited: A, B, C, E, G



DFS Traversal

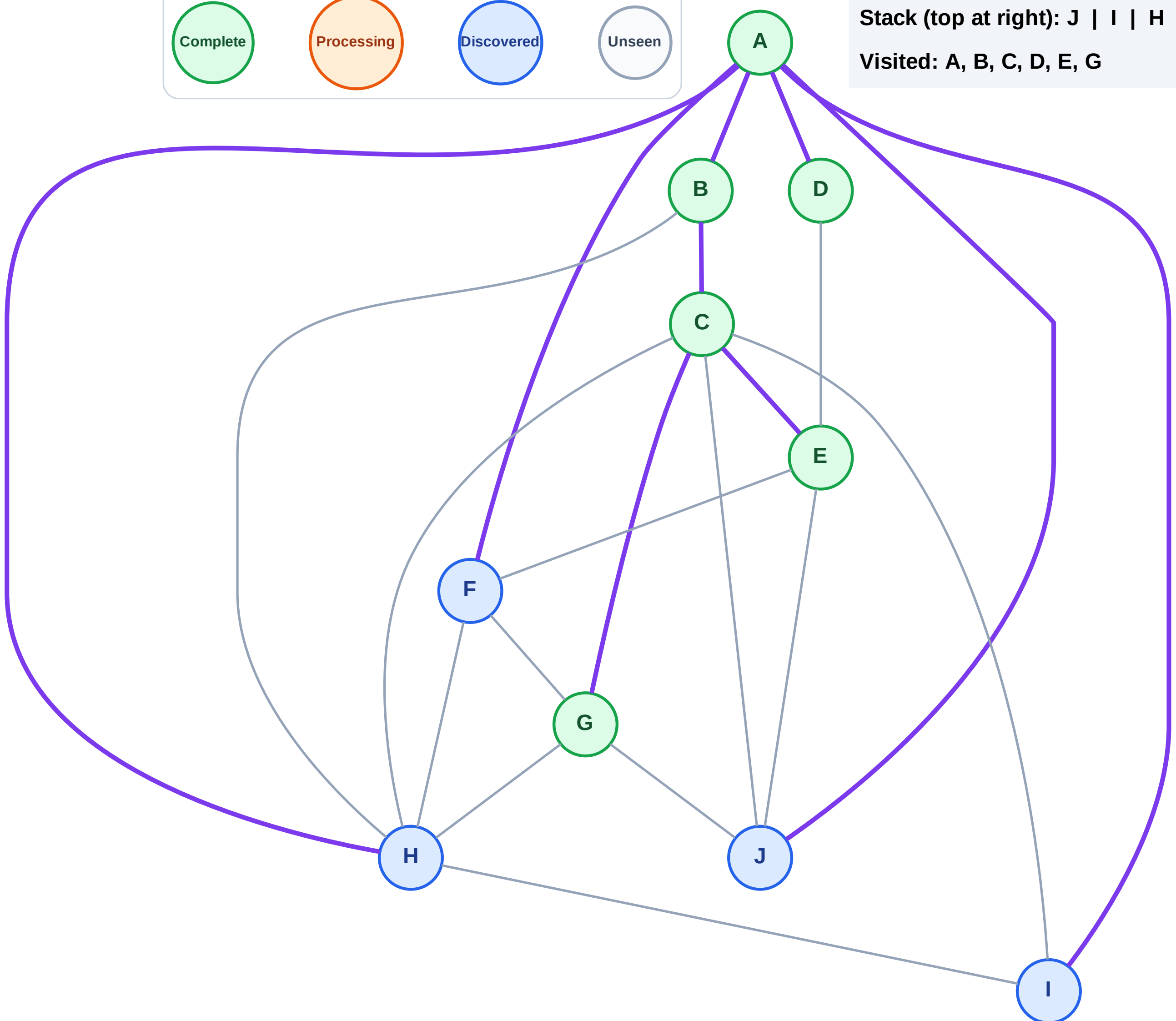
Step 13: No new neighbors from D

Legend



Stack (top at right): J | I | H | F

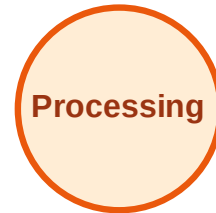
Visited: A, B, C, D, E, G



DFS Traversal

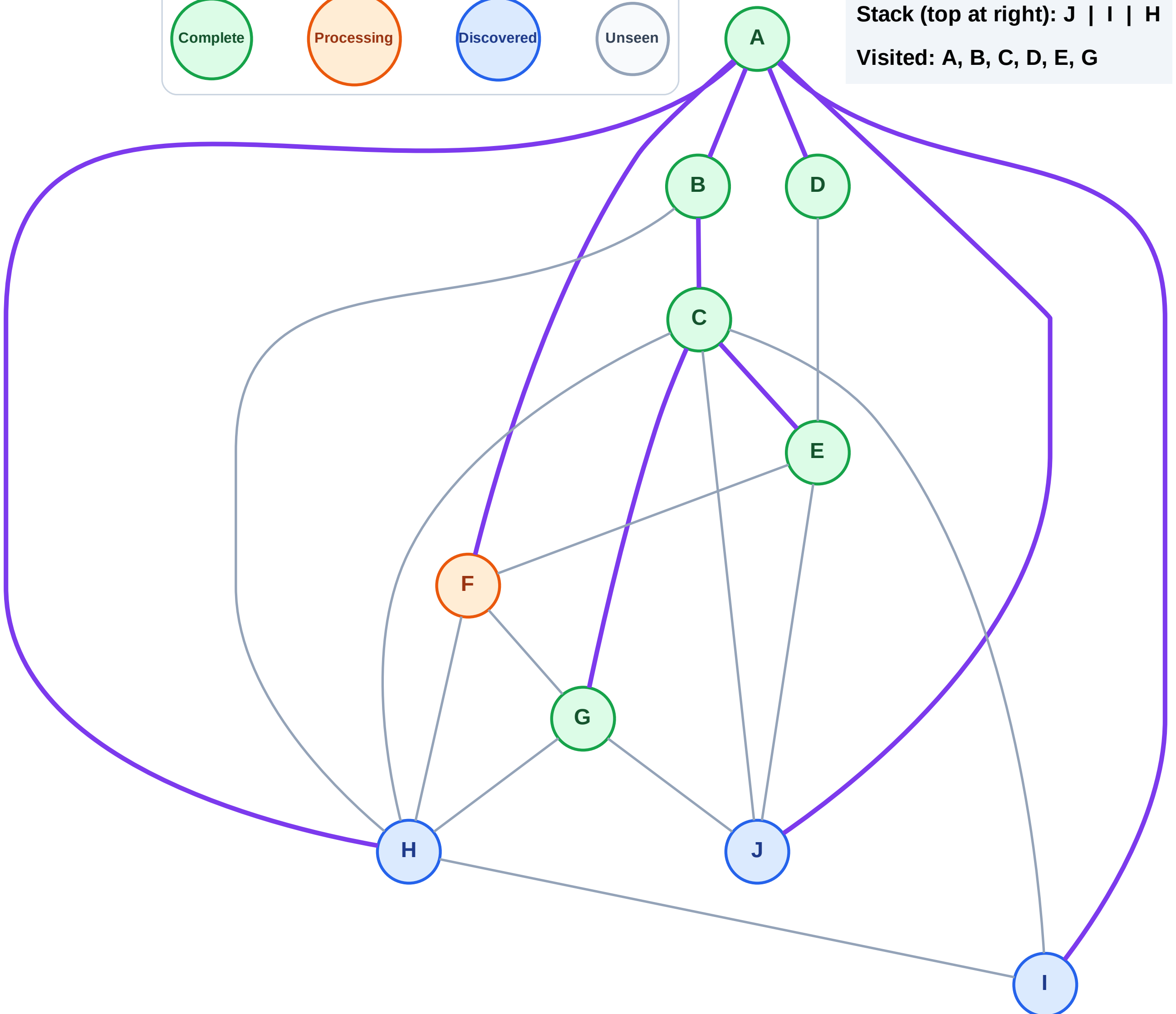
Step 14: Process node F

Legend



Stack (top at right): J | I | H

Visited: A, B, C, D, E, G



Step 15: No new neighbors from F

Step 15: No new neighbors from F

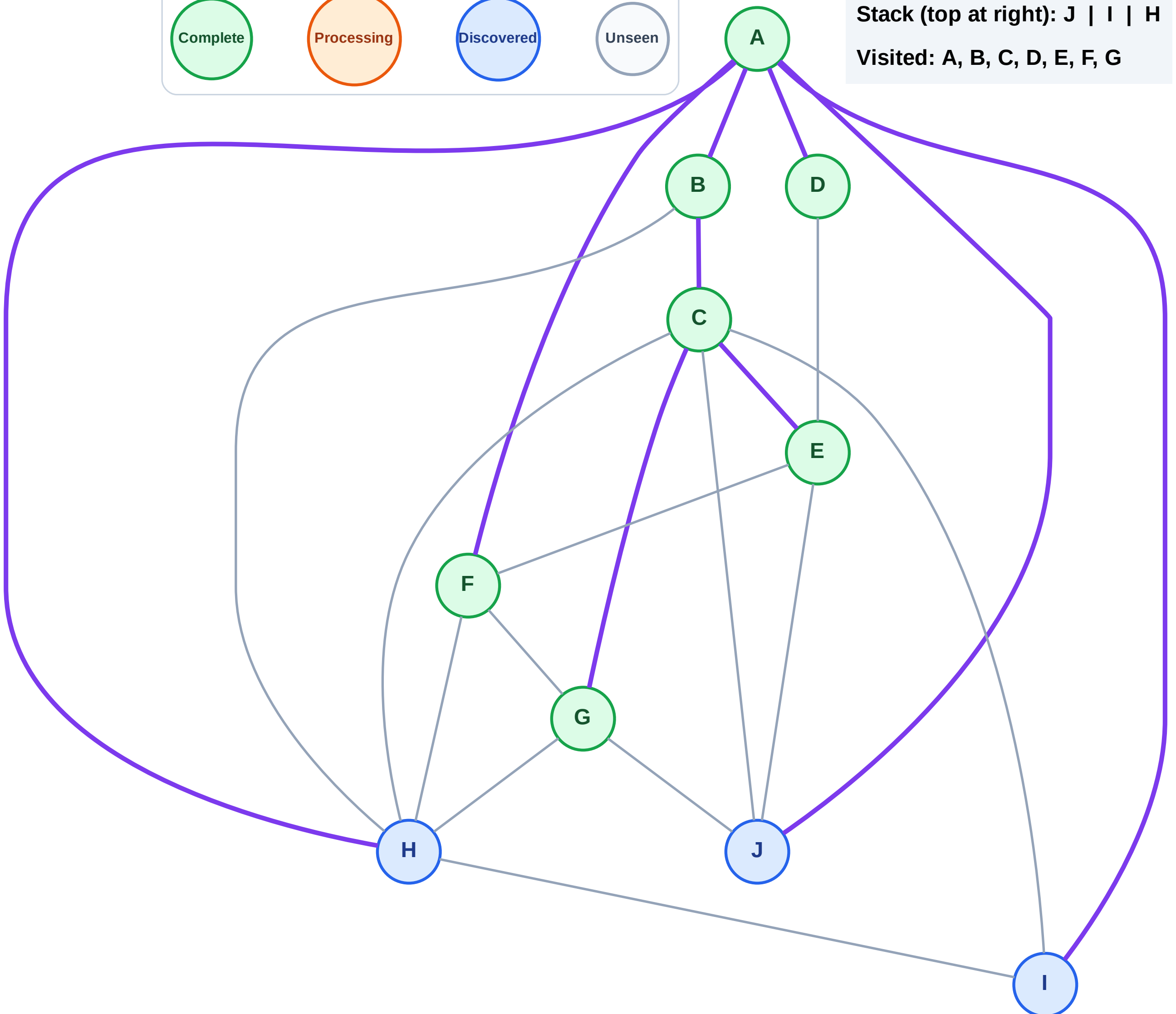
Legend

Complete Processing Discovered Unseen

Processing

Unseen

Visited: A, B, C, D, E, F, G



DFS Traversal

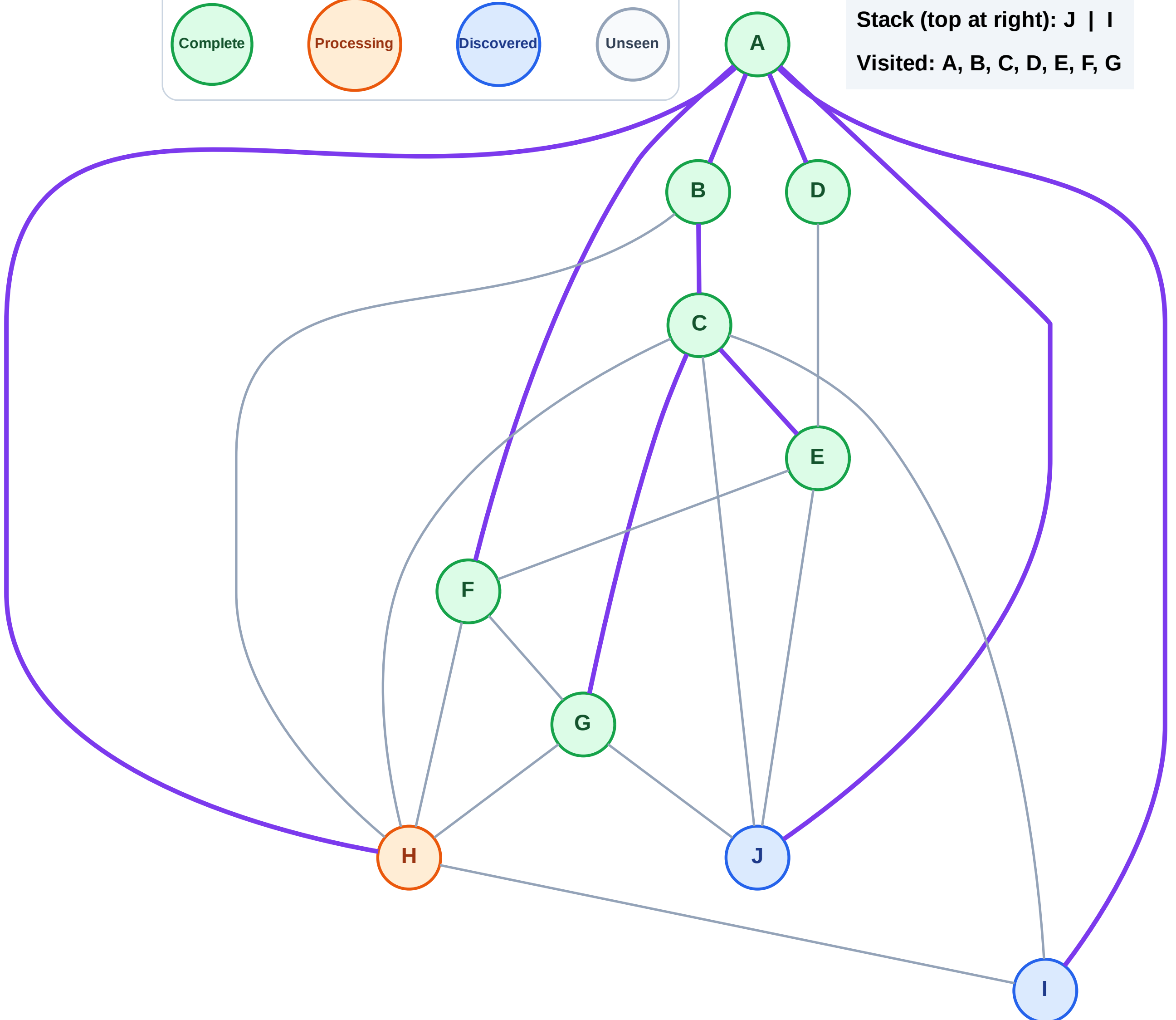
Step 16: Process node H

Legend



Stack (top at right): J | I

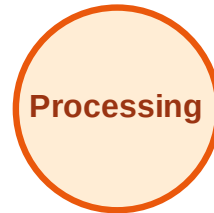
Visited: A, B, C, D, E, F, G



DFS Traversal

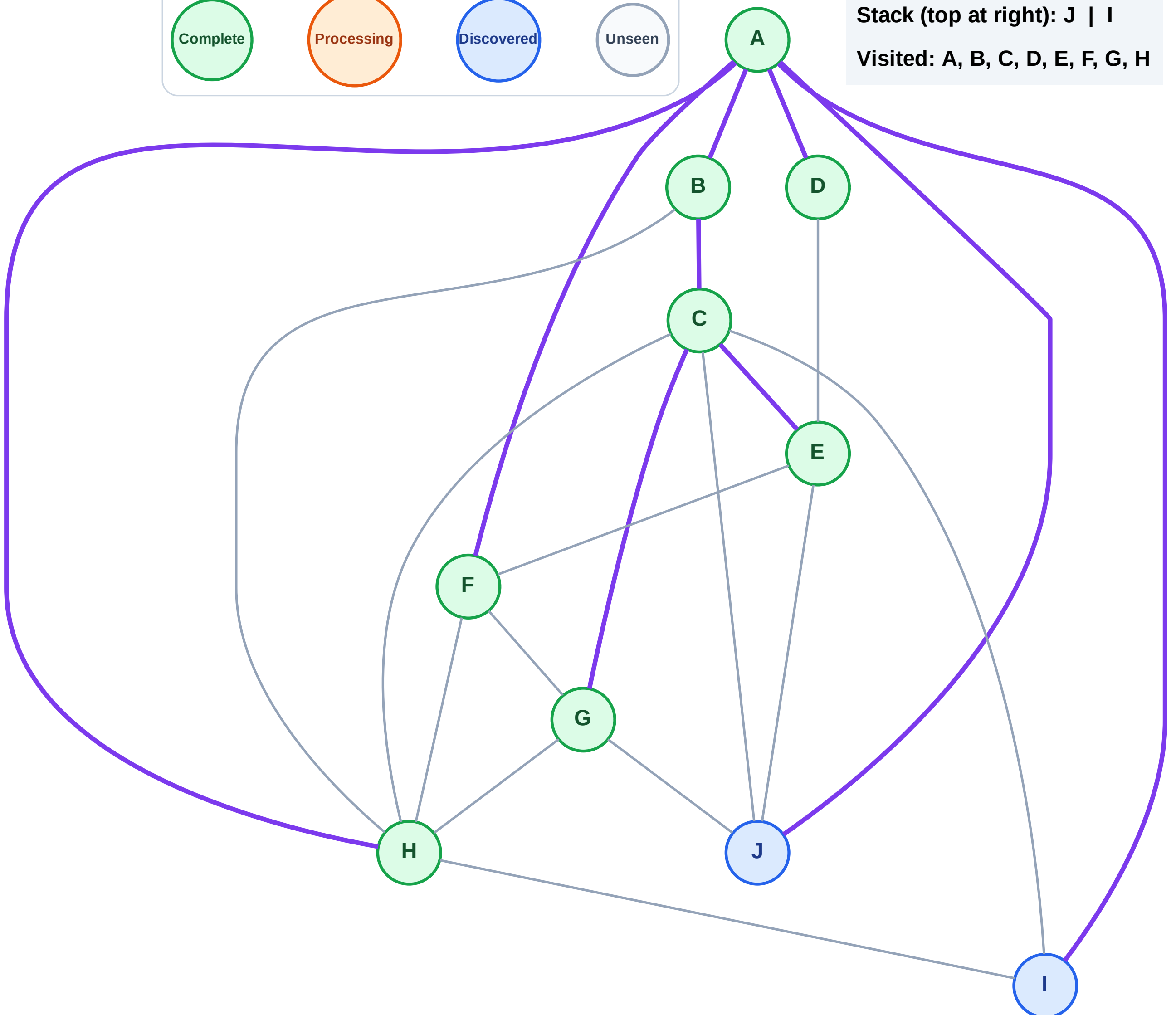
Step 17: No new neighbors from H

Legend



Stack (top at right): J | I

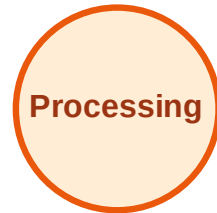
Visited: A, B, C, D, E, F, G, H



DFS Traversal

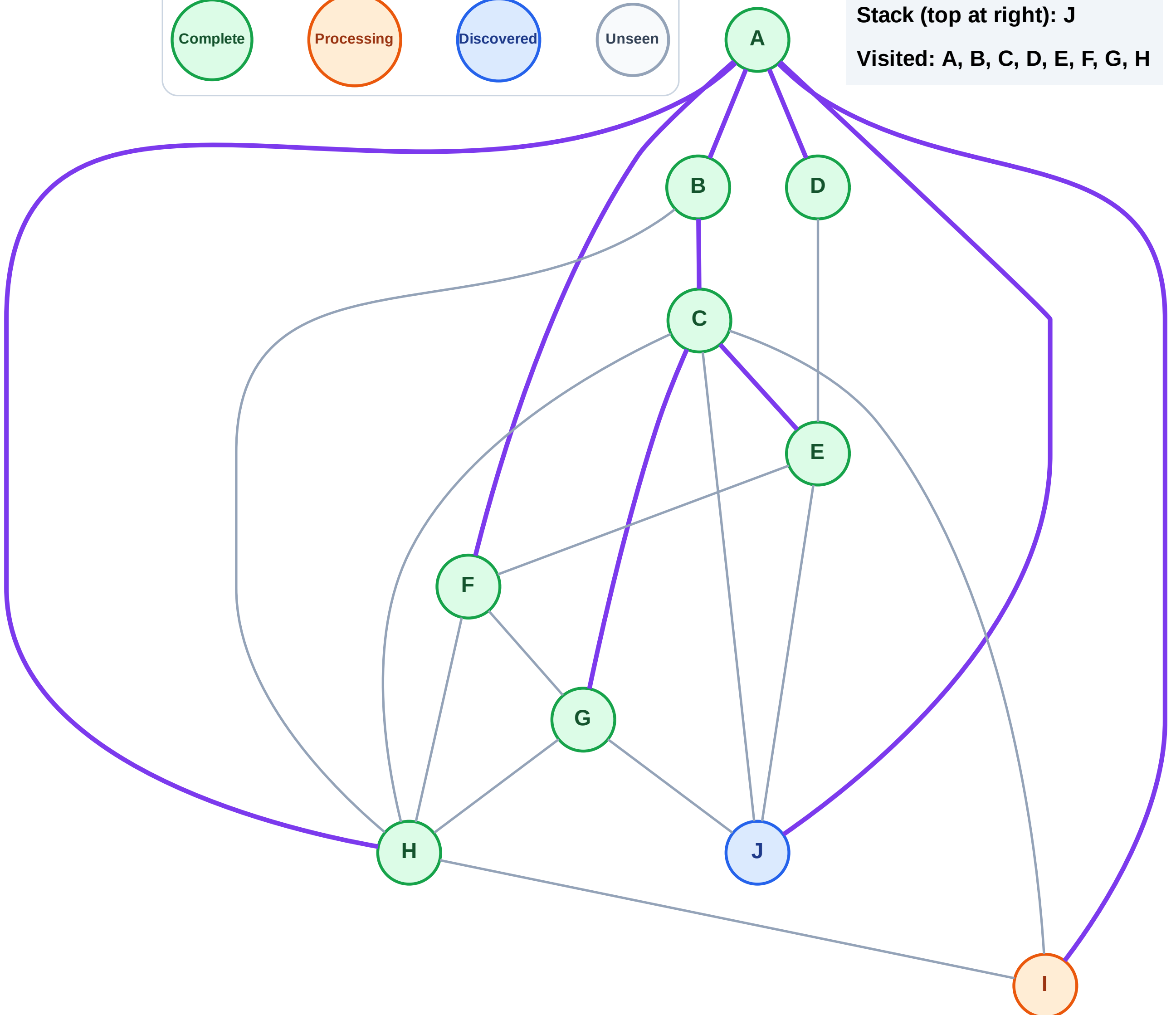
Step 18: Process node I

Legend



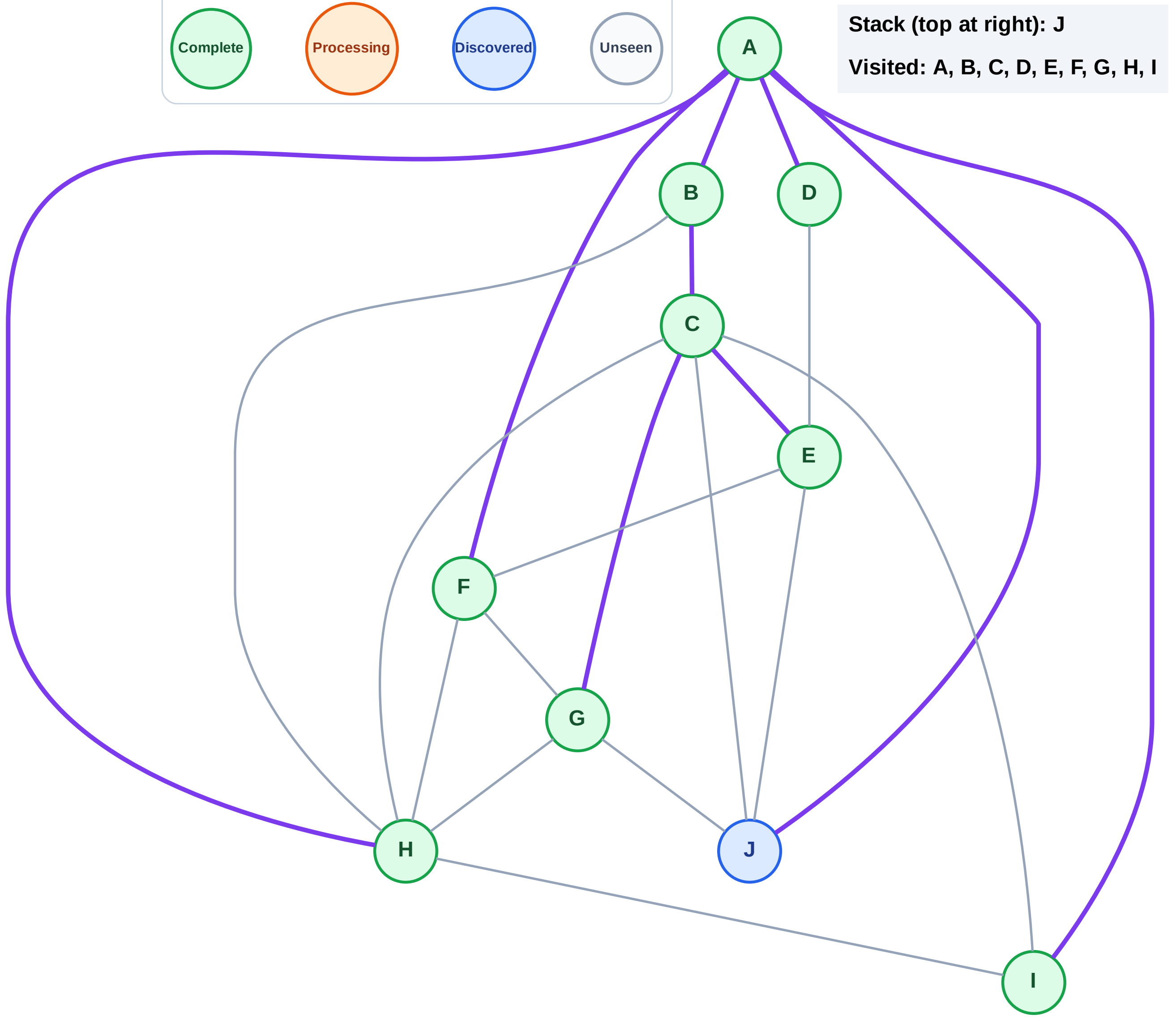
Stack (top at right): J

Visited: A, B, C, D, E, F, G, H



Step 19: No new neighbors from I

Step 19: No new neighbors from I

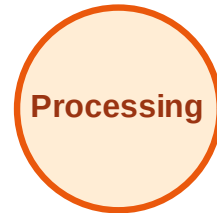


Visited: A, B, C, D, E, F, G, H, I

DFS Traversal

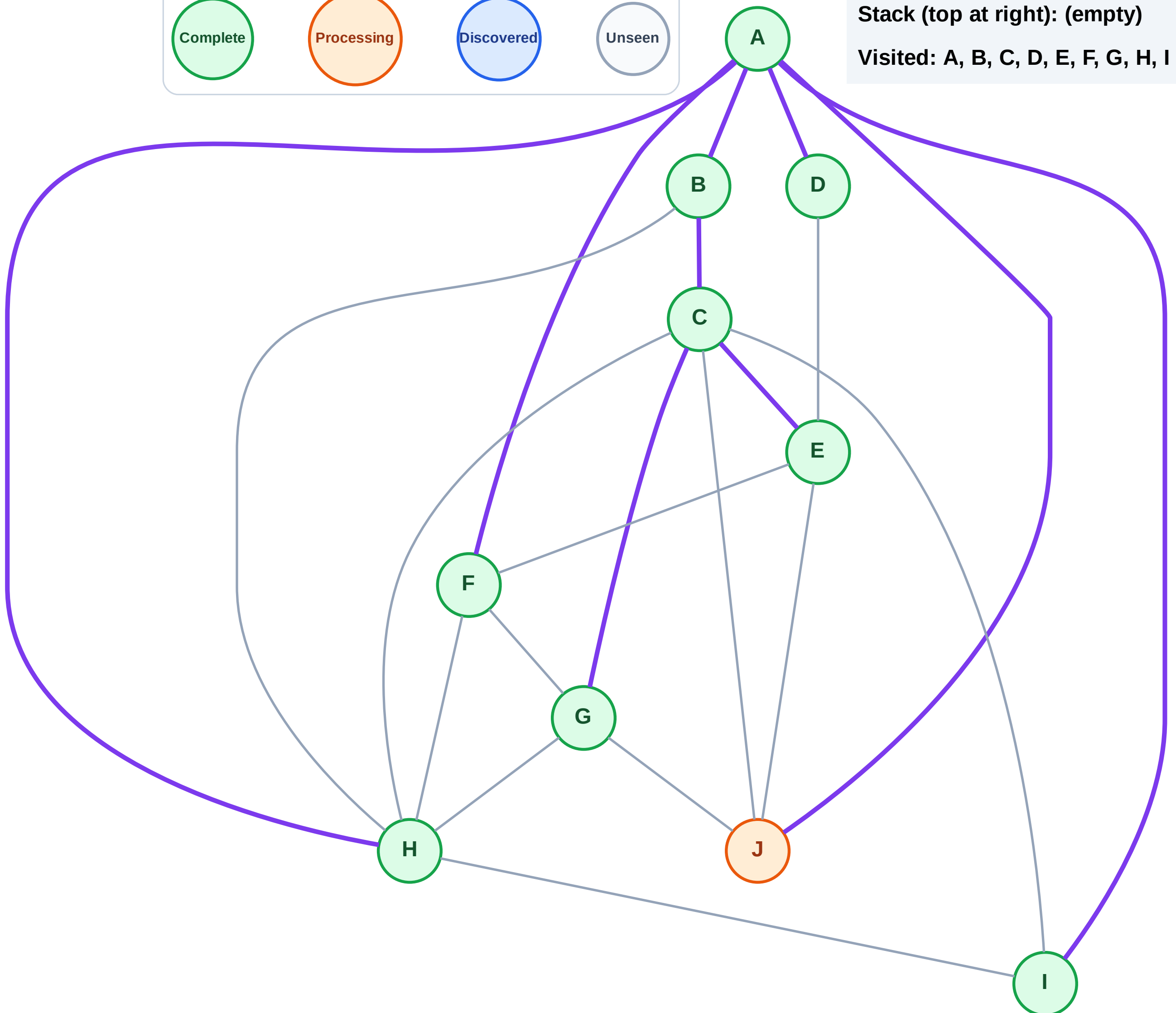
Step 20: Process node J

Legend



Stack (top at right): (empty)

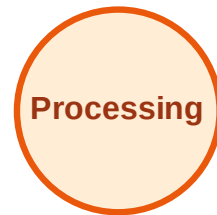
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

Step 21: No new neighbors from J

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

